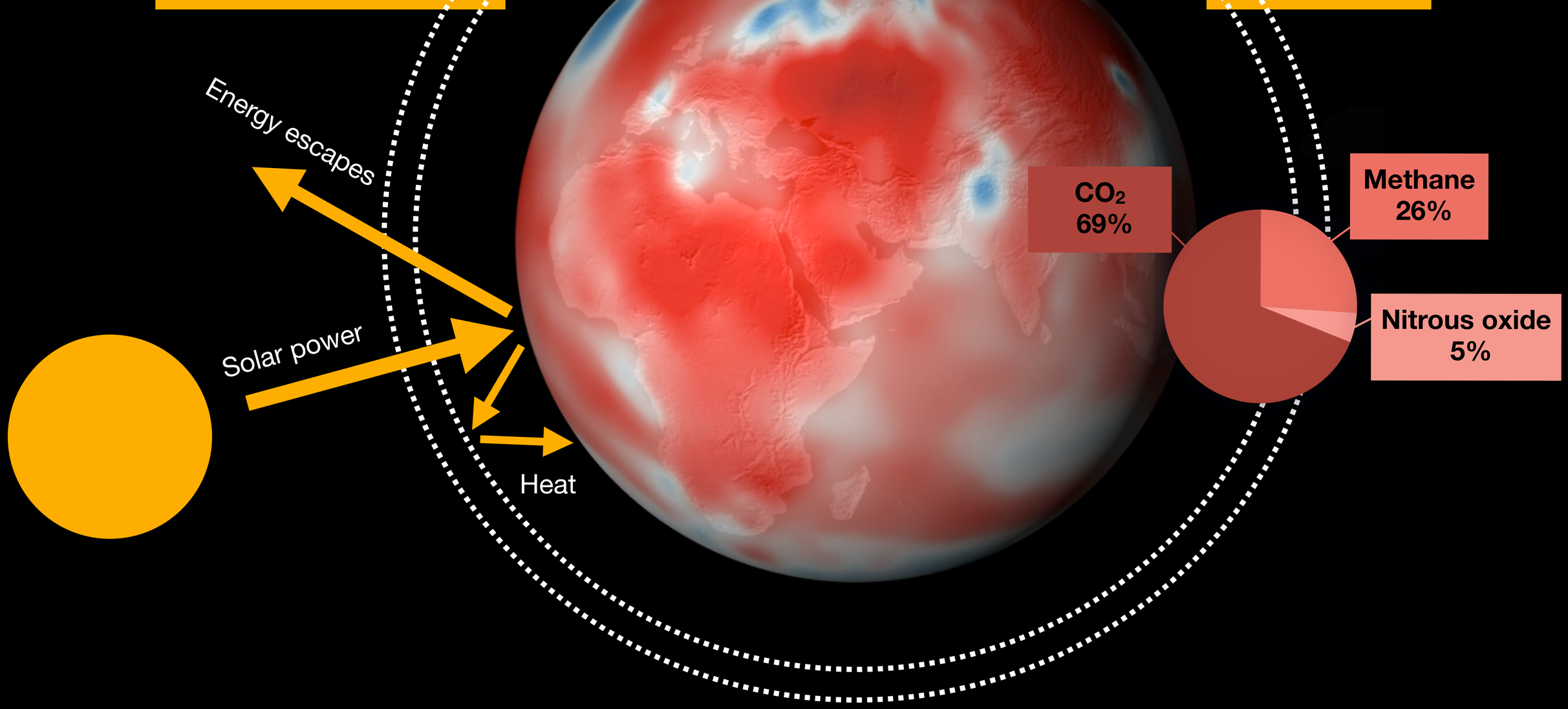
Session 1: Foundations of Marine Macroecology and Niche Theory

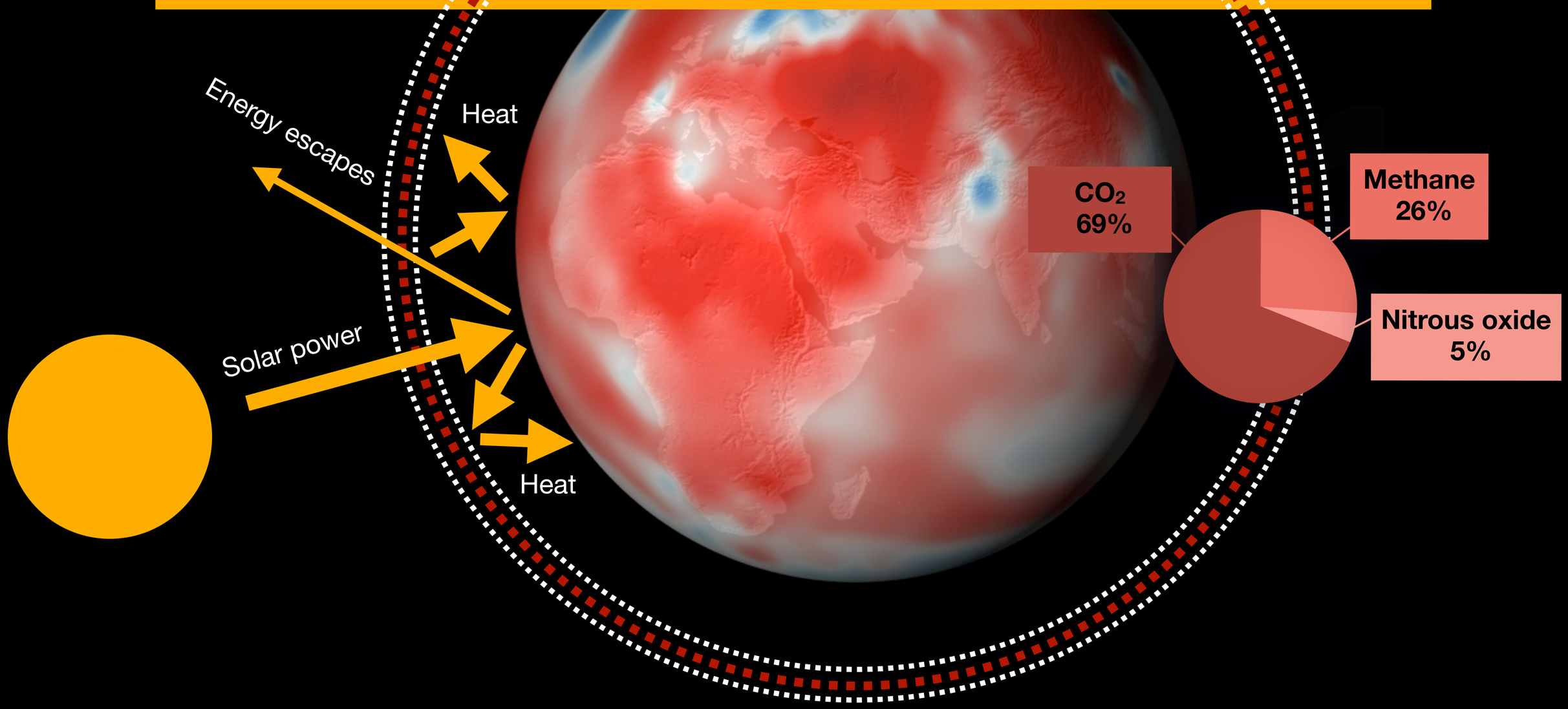
Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

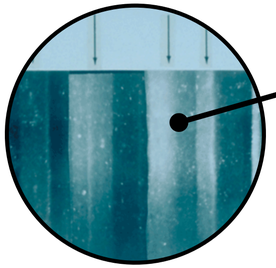
The Urgency of Modeling Biodiversity under Global Change



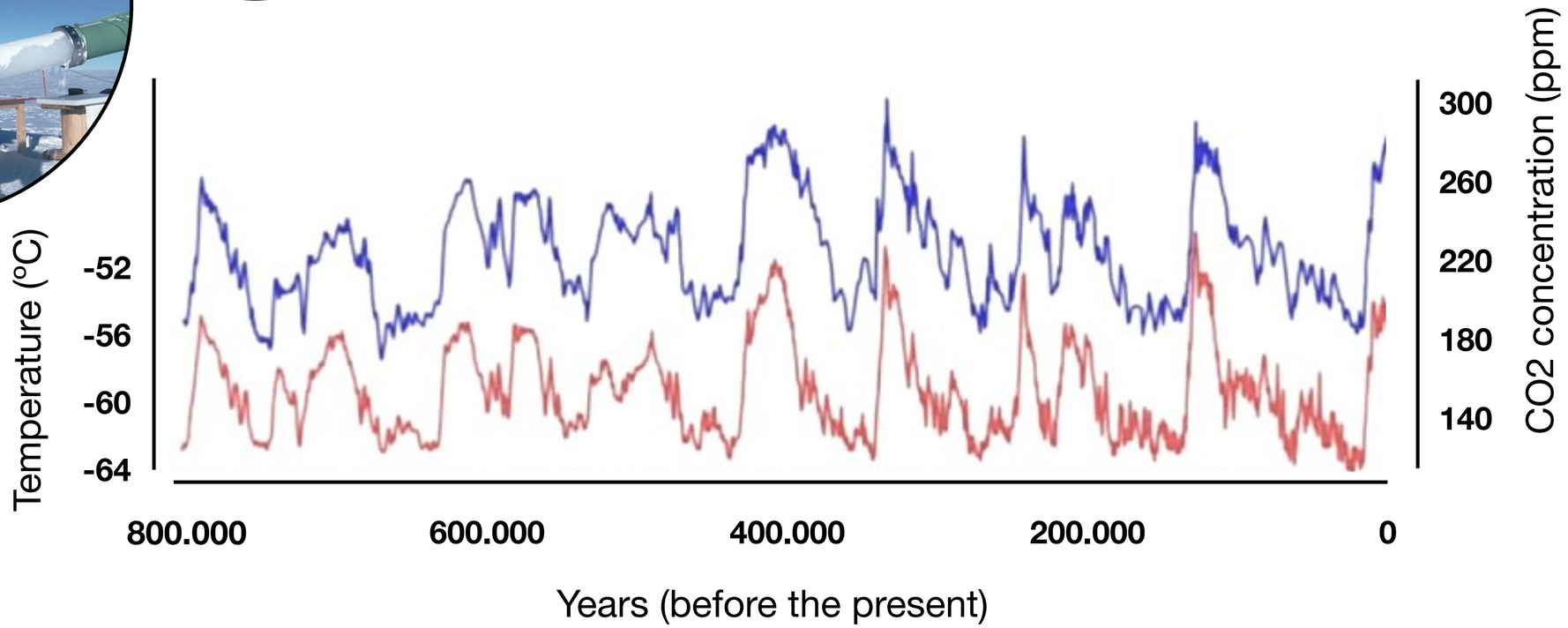
Global temperatures are linked to the concentration of **greenhouse gases**. They **absorb solar radiation, reducing heat loss into space**, thus controlling the global temperature.



The **higher the concentration of greenhouse gases** in the atmosphere, the **greater the heat retention** on the Earth's surface, leading to global warming.

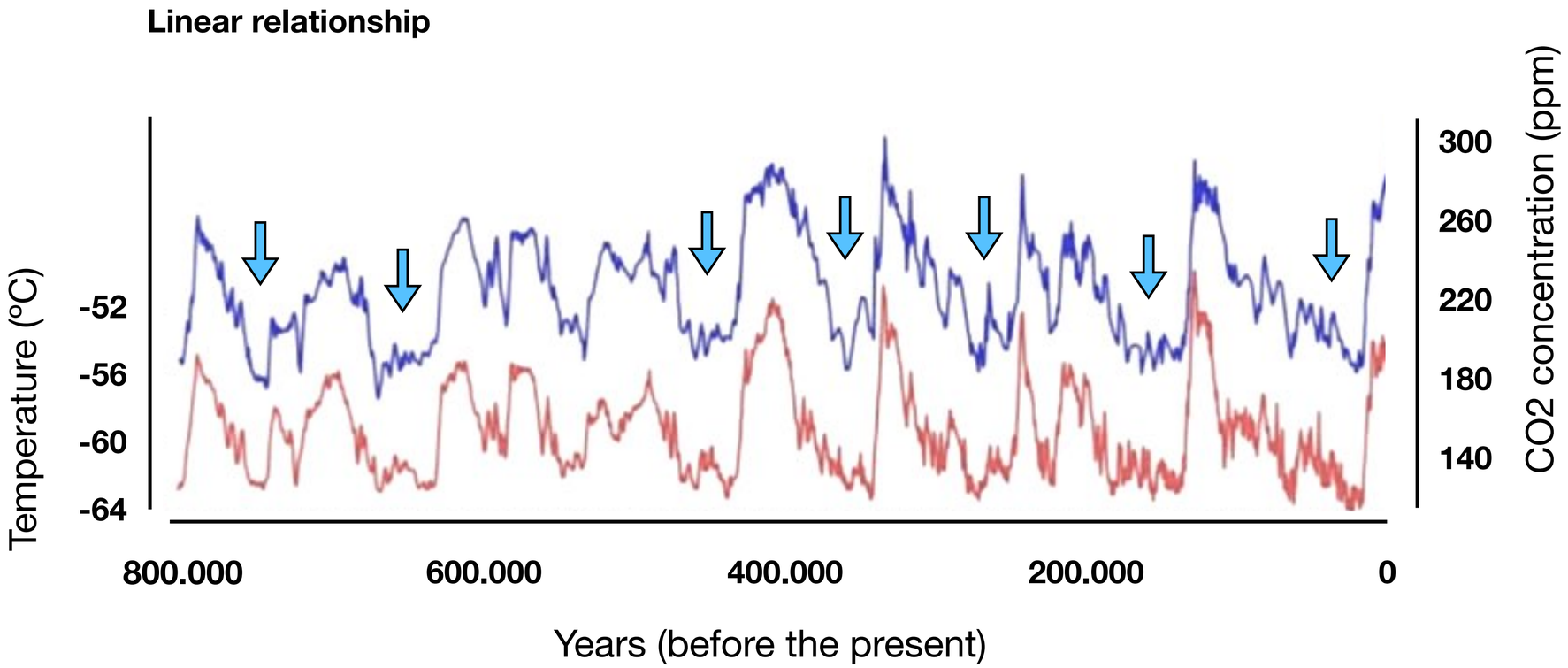


Investigators drill the ice
Antarctica



The relationship between greenhouse gases and temperature is well studied.

Ice traps **atmospheric gases**, which allow to **reconstruct the concentrations of greenhouse gases** and **estimate the earth's temperature** (indicators such as *oxygen isotopes*).



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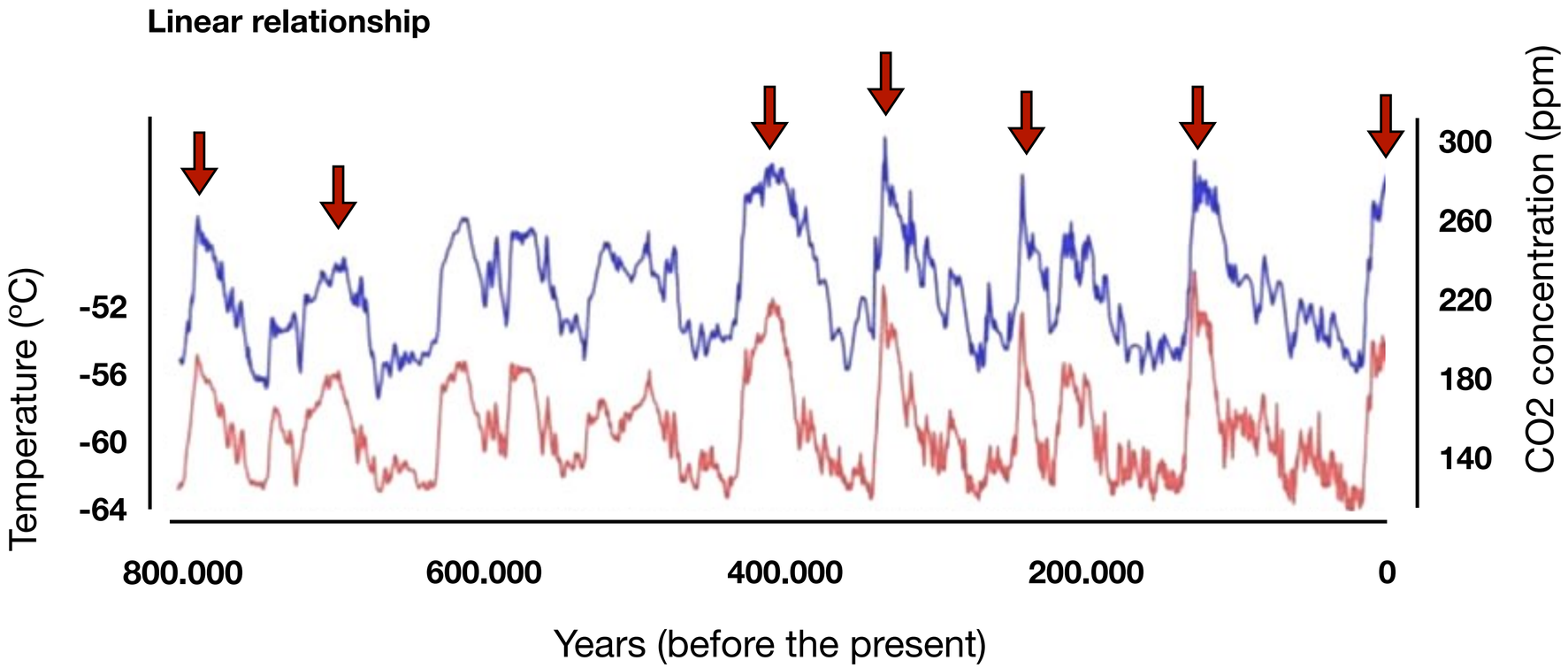

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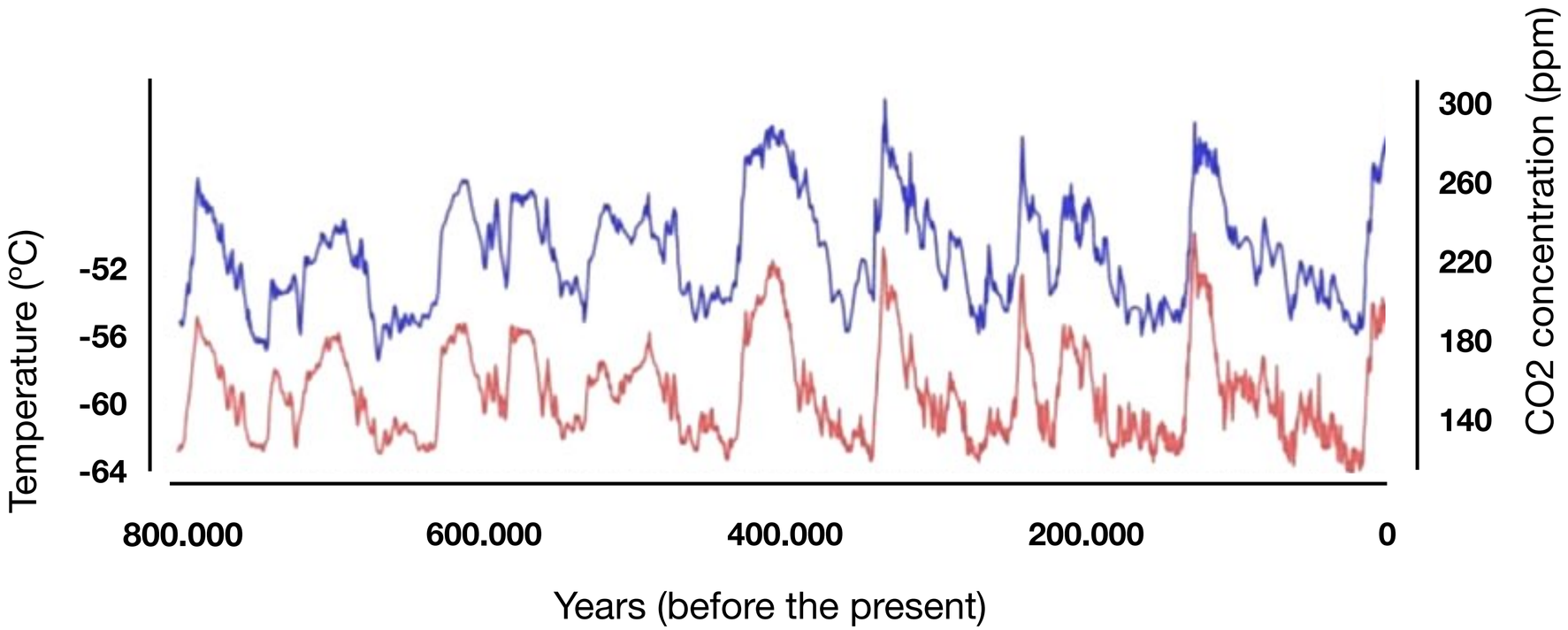

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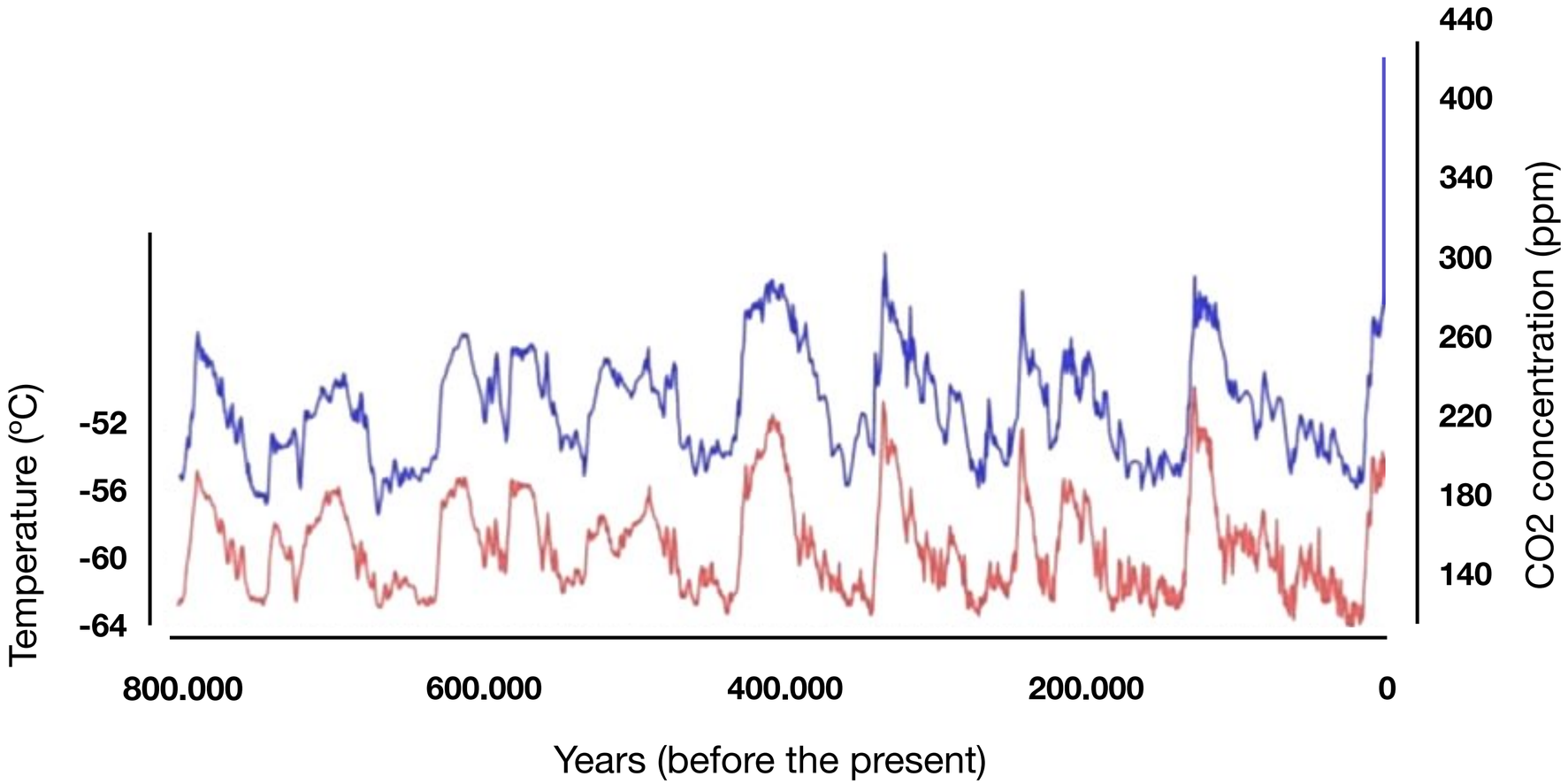
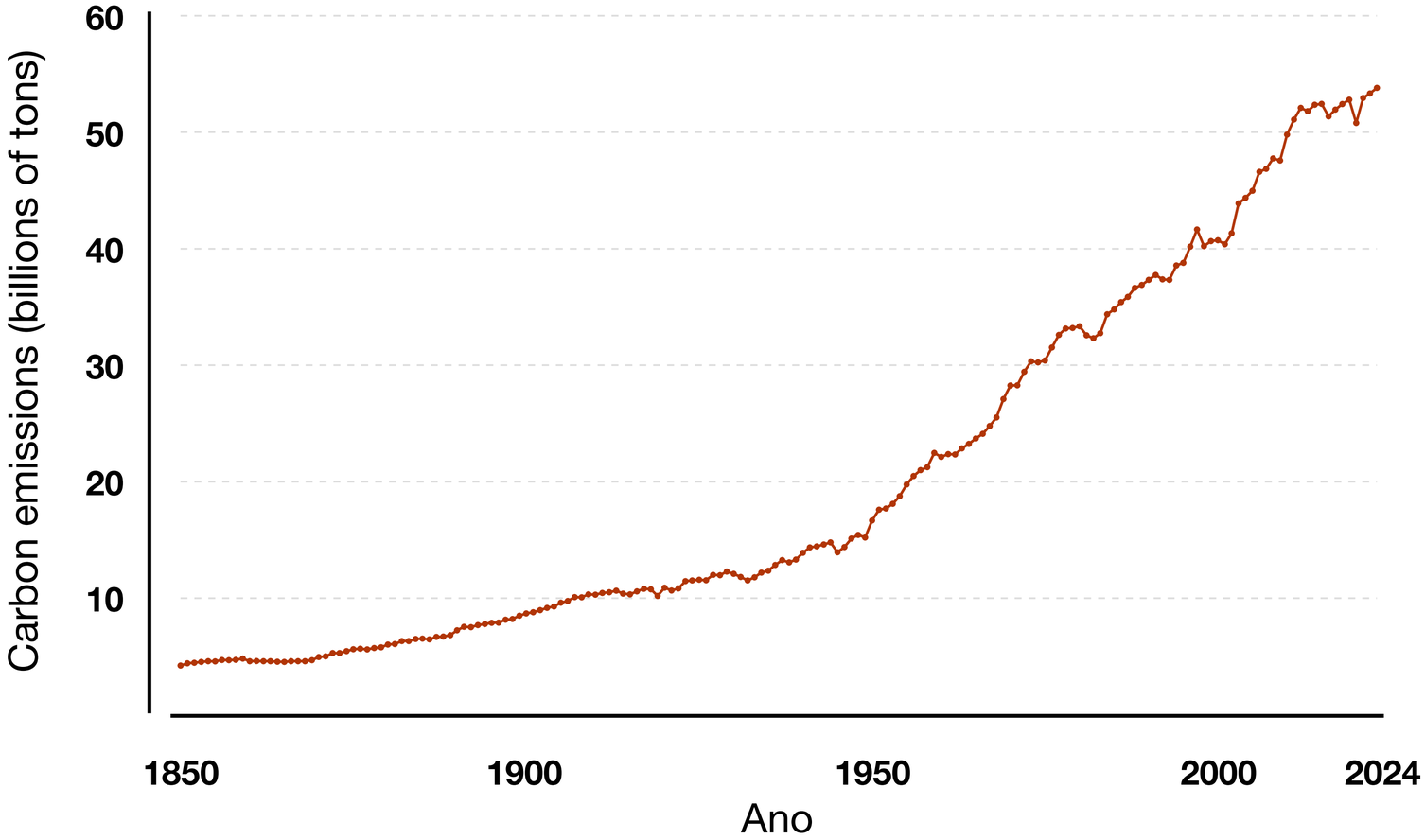
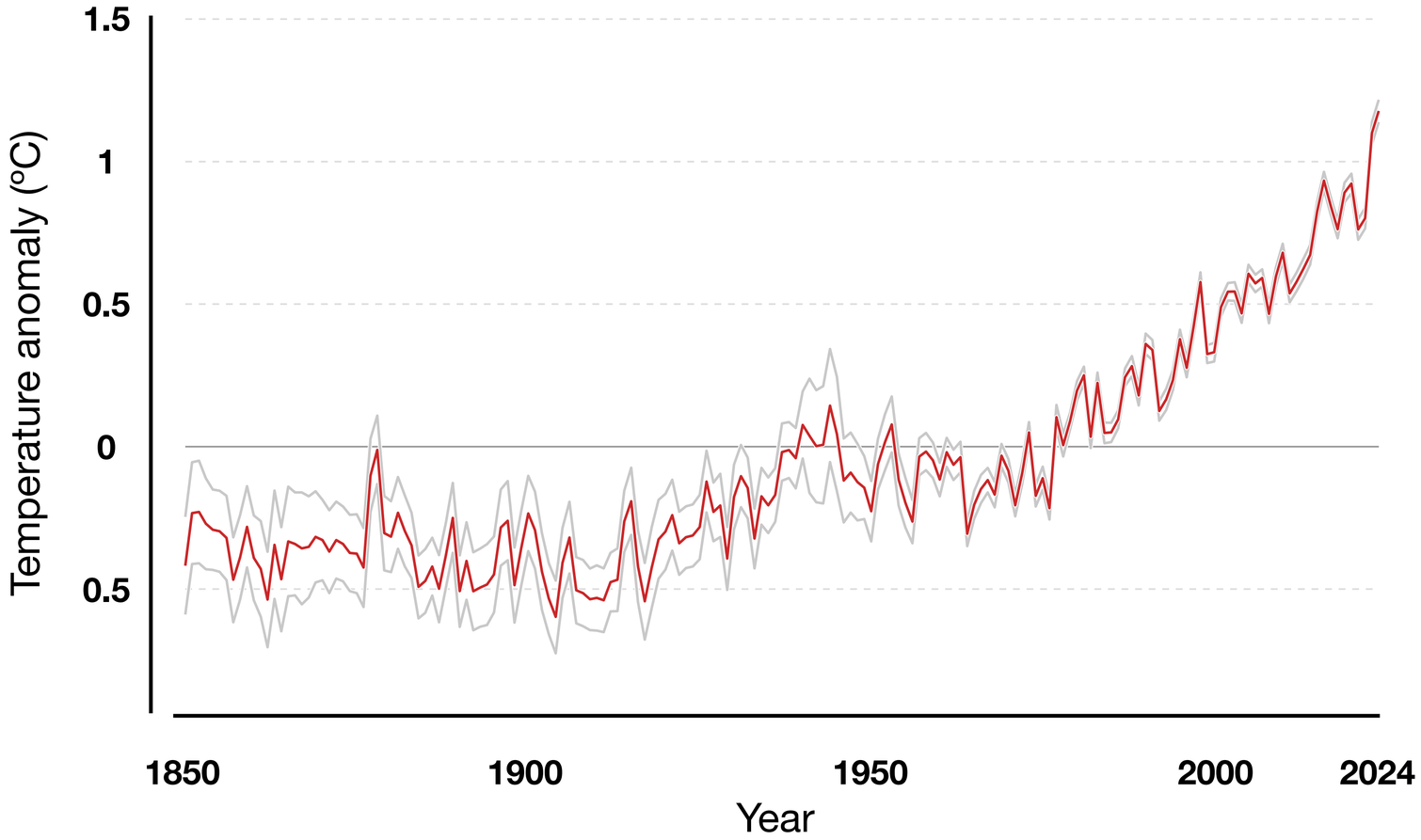


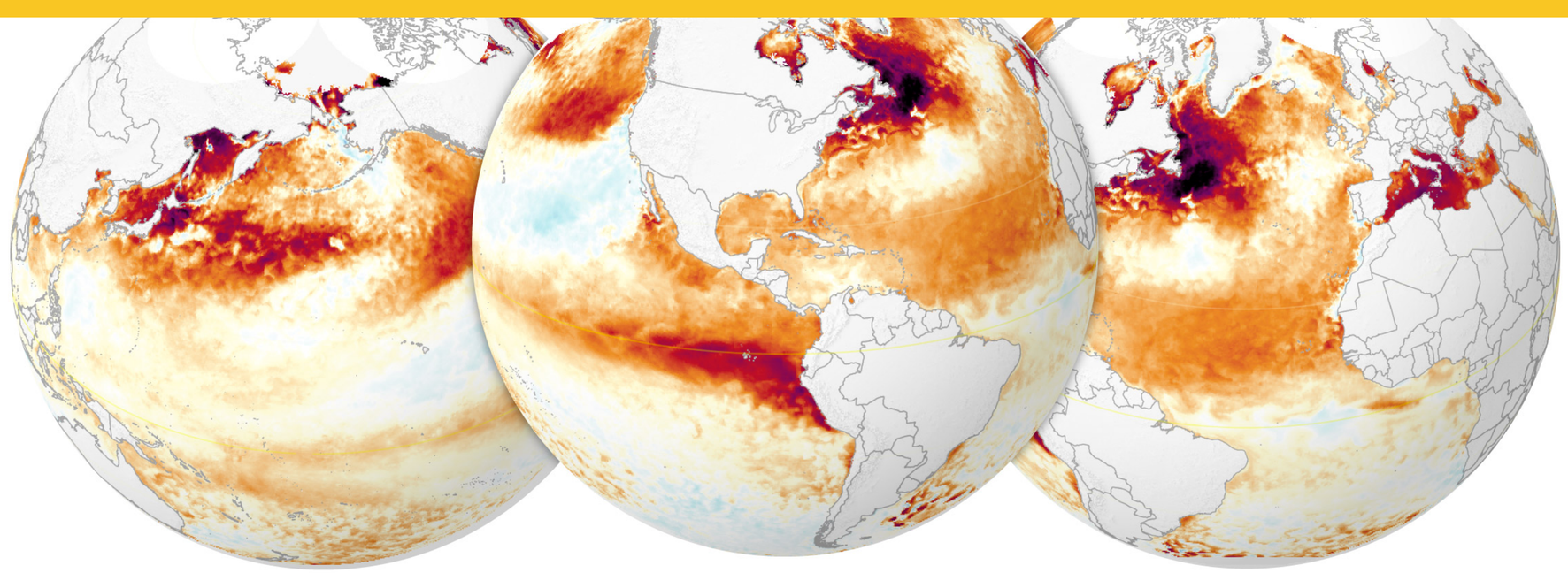
Since the **beginning of the Industrial Revolution**, the rate of **increase in greenhouse gases is 100 times higher than any natural oscillation** recorded in the last 800,000 years.



The **last time** GHG were **so high was 3 million years ago** (mid-Pliocene): Closest natural analogue for today's CO_2 levels, a period ~2–4 °C warmer than preindustrial, with high latitudes (e.g., Arctic) up to +10 °C and absent permanent sea ice. Sea levels were ~10–25 meters higher.



With the increased concentrations of GHG since pre-industrial, **temperatures increased by about 1.4°C**. In the past, the **natural increase of 1°C took thousands of years**; now, just over a century.



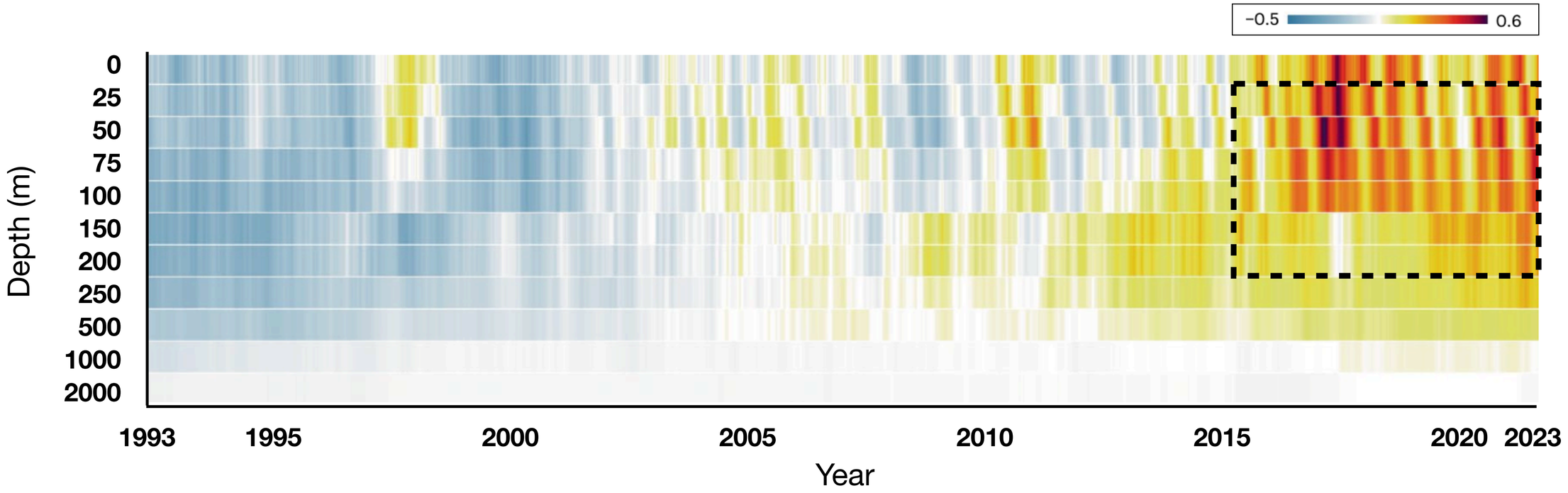
-ºC +ºC

Heat waves in the ocean

Increased concentrations of GHG aggravate “marine heatwaves" - **periods of extreme ocean warming that can last for days, weeks or even months**.

MHW are now more frequent, more intense and last longer.

Once rare phenomena, they now occur several times a decade.

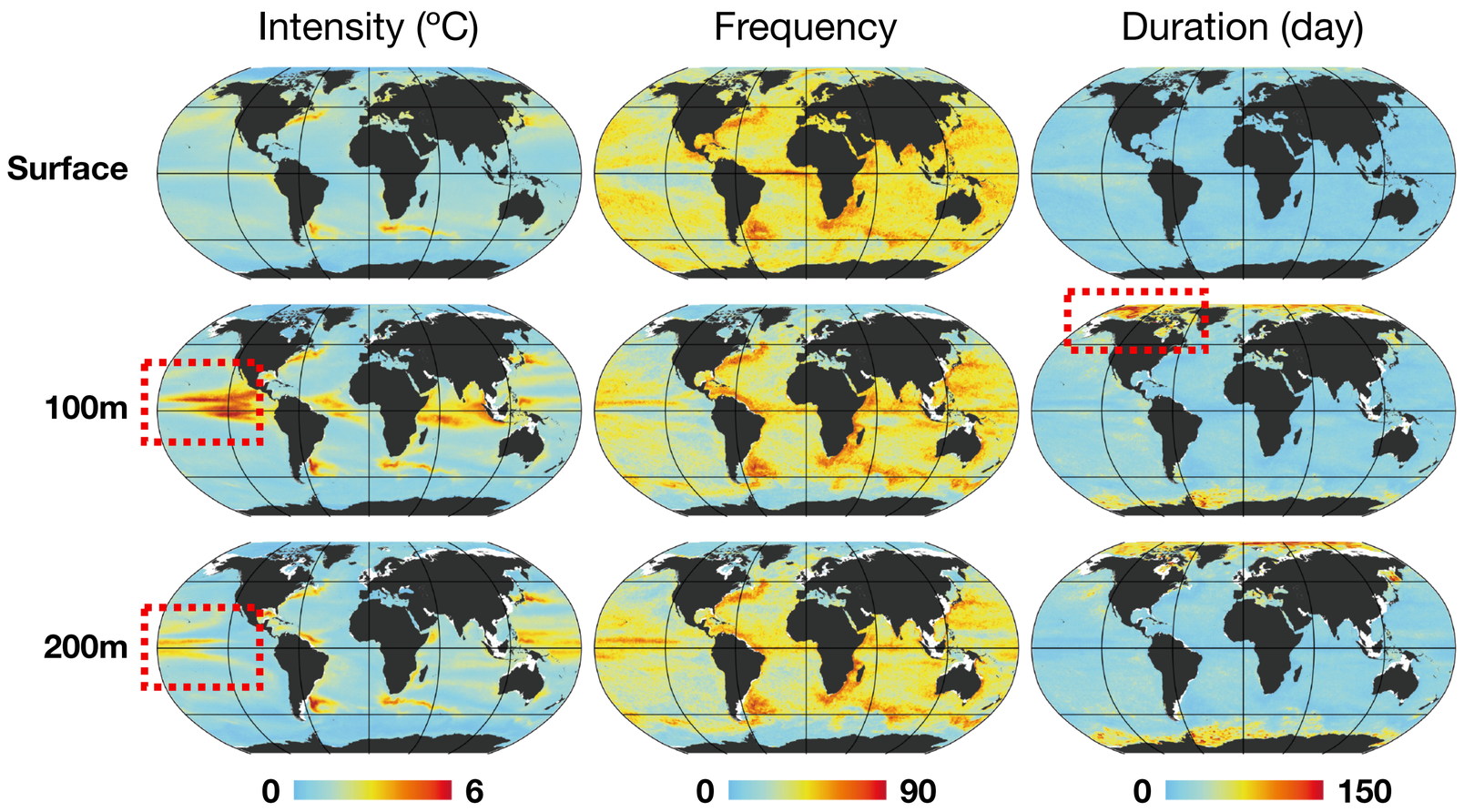


MHW were previously thought to be surface only, but they exist in depth.

We used 30 years of daily temperature data, from the surface to 2,000 meters, we showed how climate change has modified the entire ocean.

More intense MHW at 50-200 meters, and not on the surface.

Fragkopoulou et al, 2023



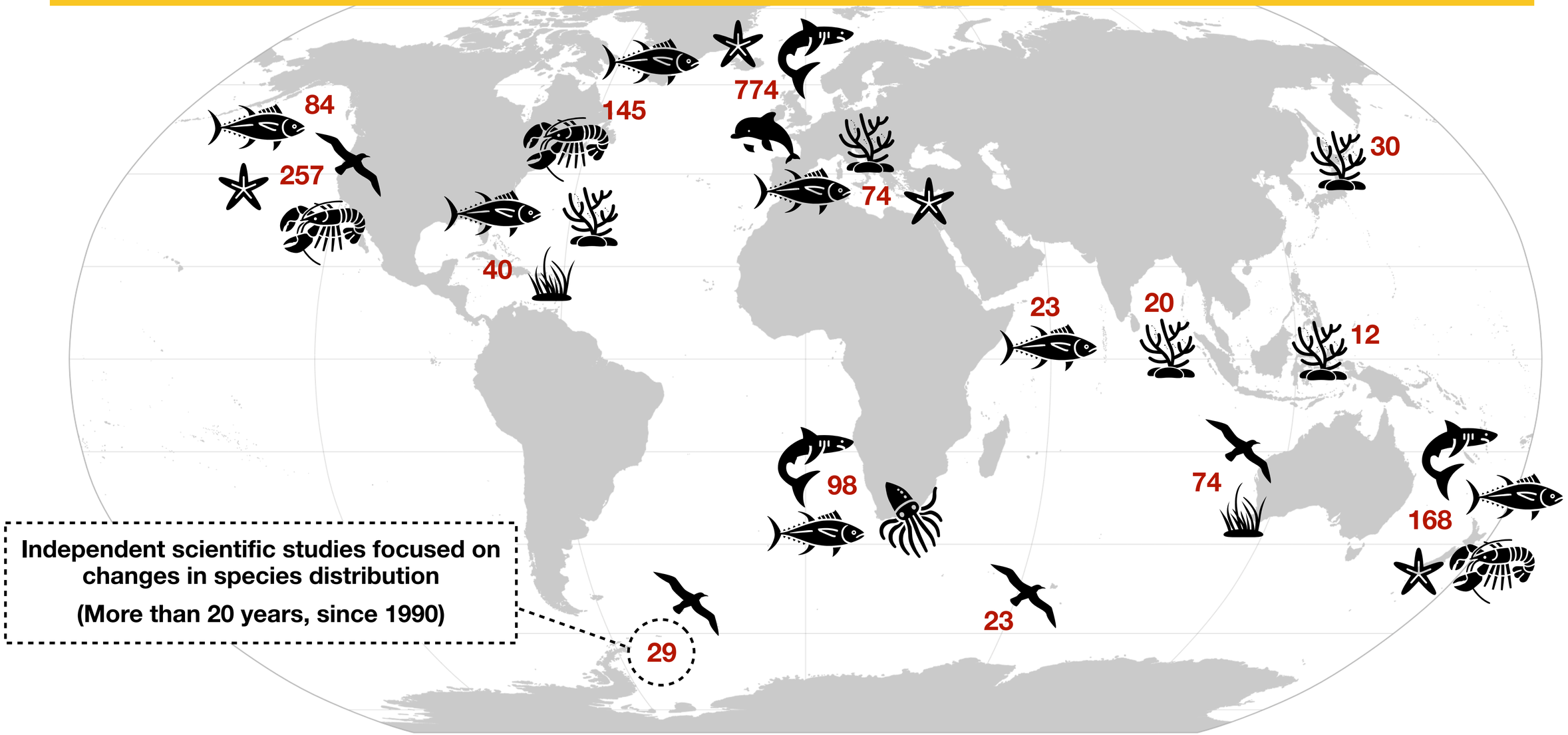
Deeper regions of the sea (e.g., 100 to 200m) exhibit **extreme MHW of up to 6ºC**, **very frequent** (up to 90 times in the last 30 years) **and long-lasting** (Arctic, MHW with duration > 150 days).

Fragkopoulou et al, 2023

The **increase in greenhouse gases has increased the temperature of the oceans**, with more intense, frequent, long-lasting MHW, also in depth.

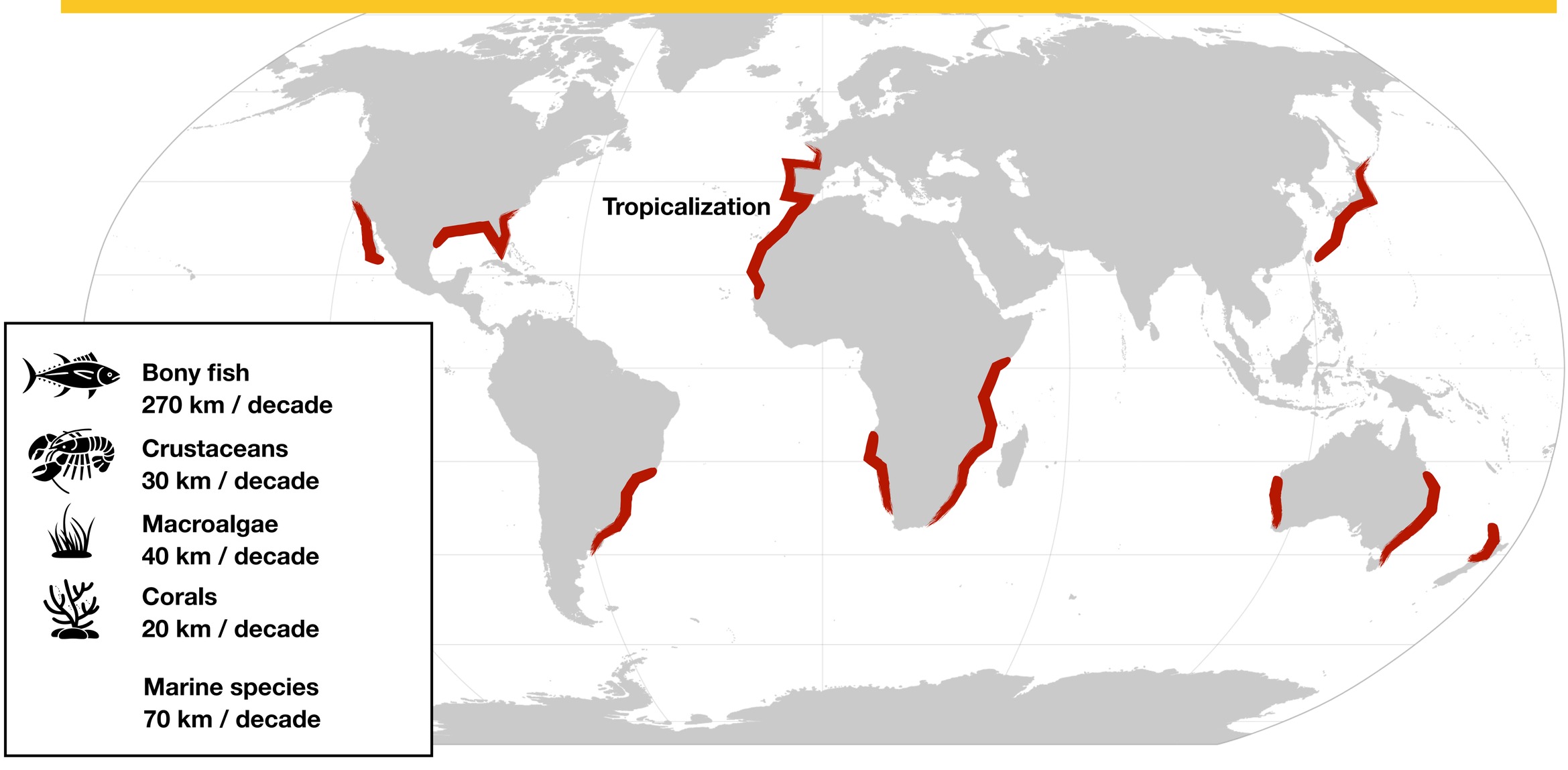
The distribution of species is fundamentally determined by temperature…

What is the impact of recent climate change on marine species?



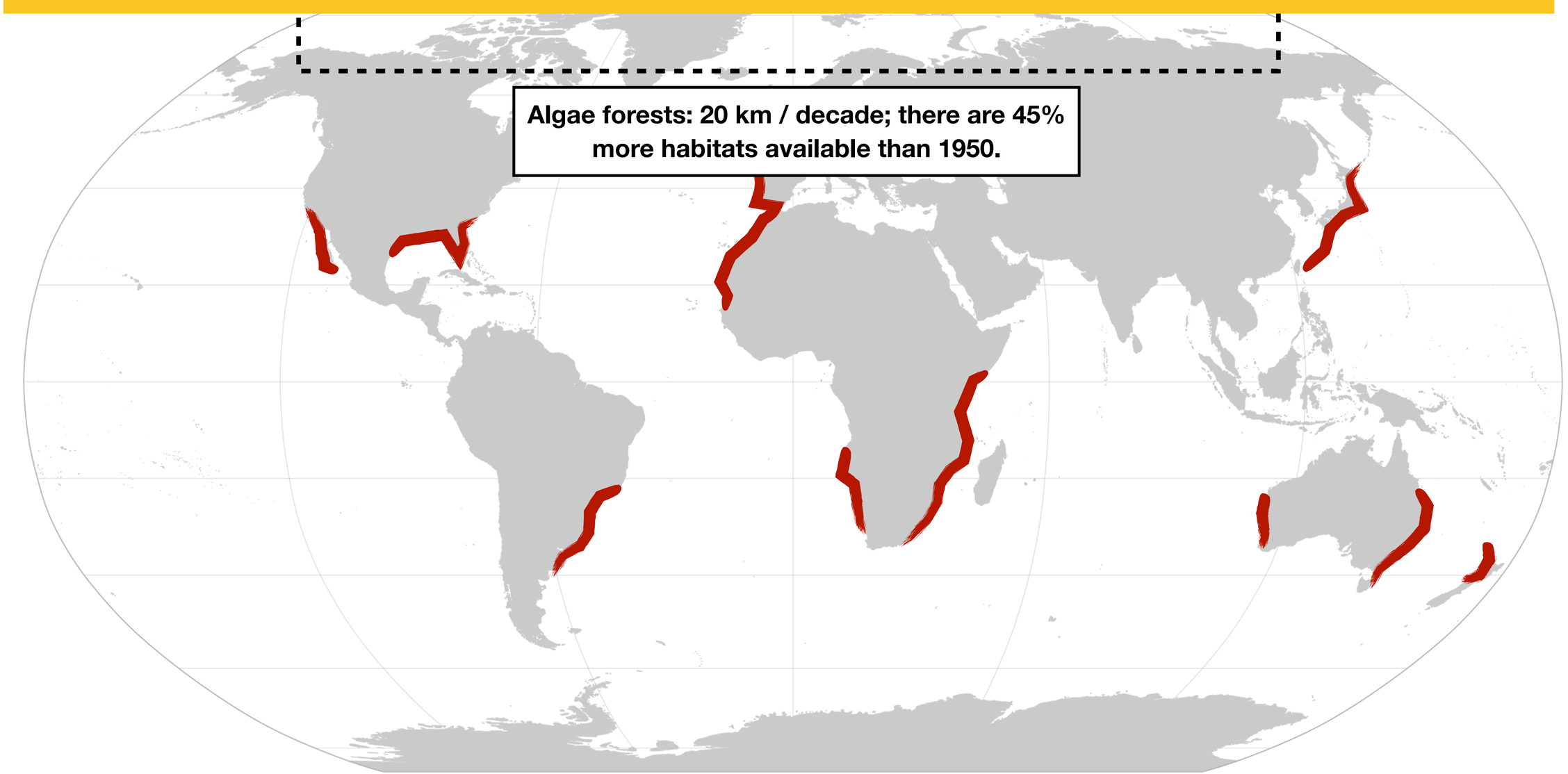
An extensive review showed ongoing climate change impacting all taxonomic / ecological groups (e.g., from corals to sharks). Changes are higher than those recorded on land.

Pinsky et al., 2020



An additional review show species shifting towards the poles (70 km / decade) - tropicalization of the oceans - traditionally cold or temperate regions exhibit now more tropical species.

Poloczanska et al., 2016



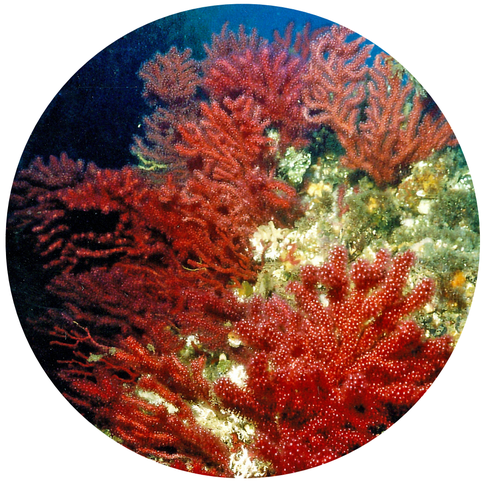
An additional review show the *borealization* of the Arctic - the process where ecosystems transform to resemble warmer, lower-latitude boreal (sub-Arctic) regions.

Assis et al., 2022

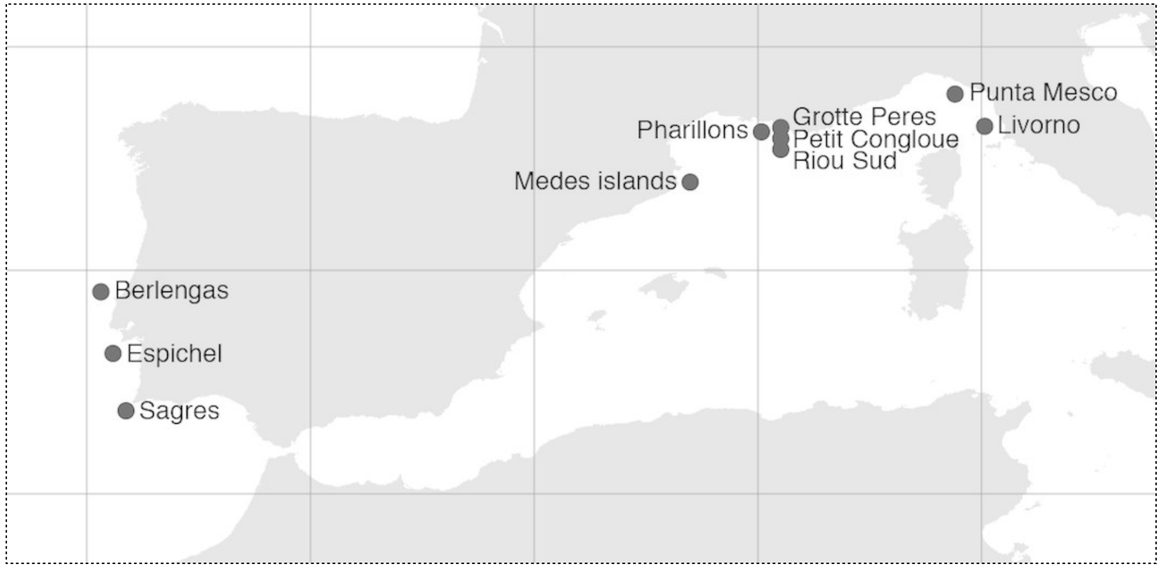
For **species able to shift distributions vertically** (altitudinal shifts), can **deep, colder waters provide refugia for biodiversity?**

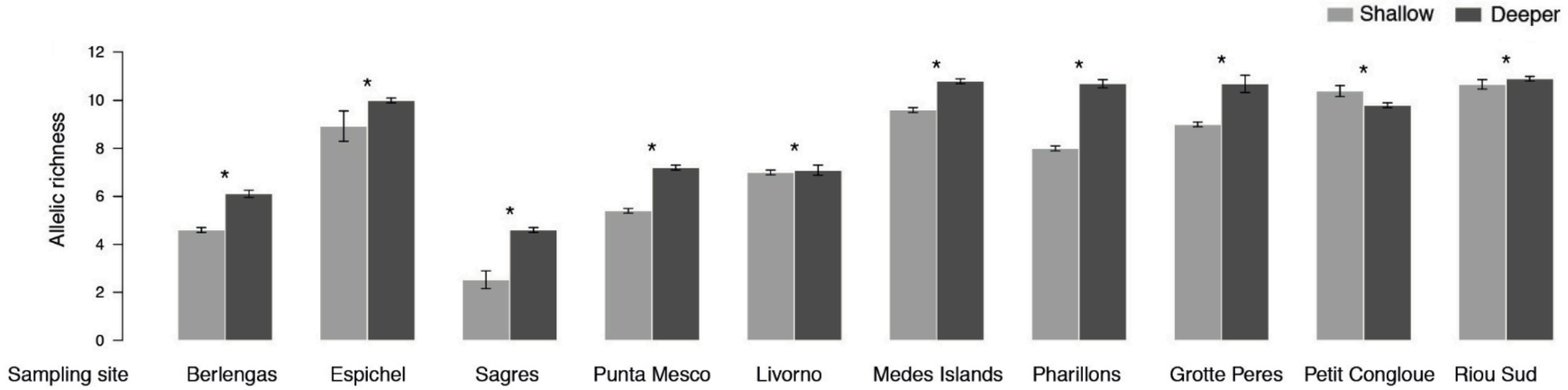
We estimated population **genetic diversity across the distribution of a coral**. We **compared the differences in diversity between shallow and deep populations at each site**.

Pilczynska et al., 2019



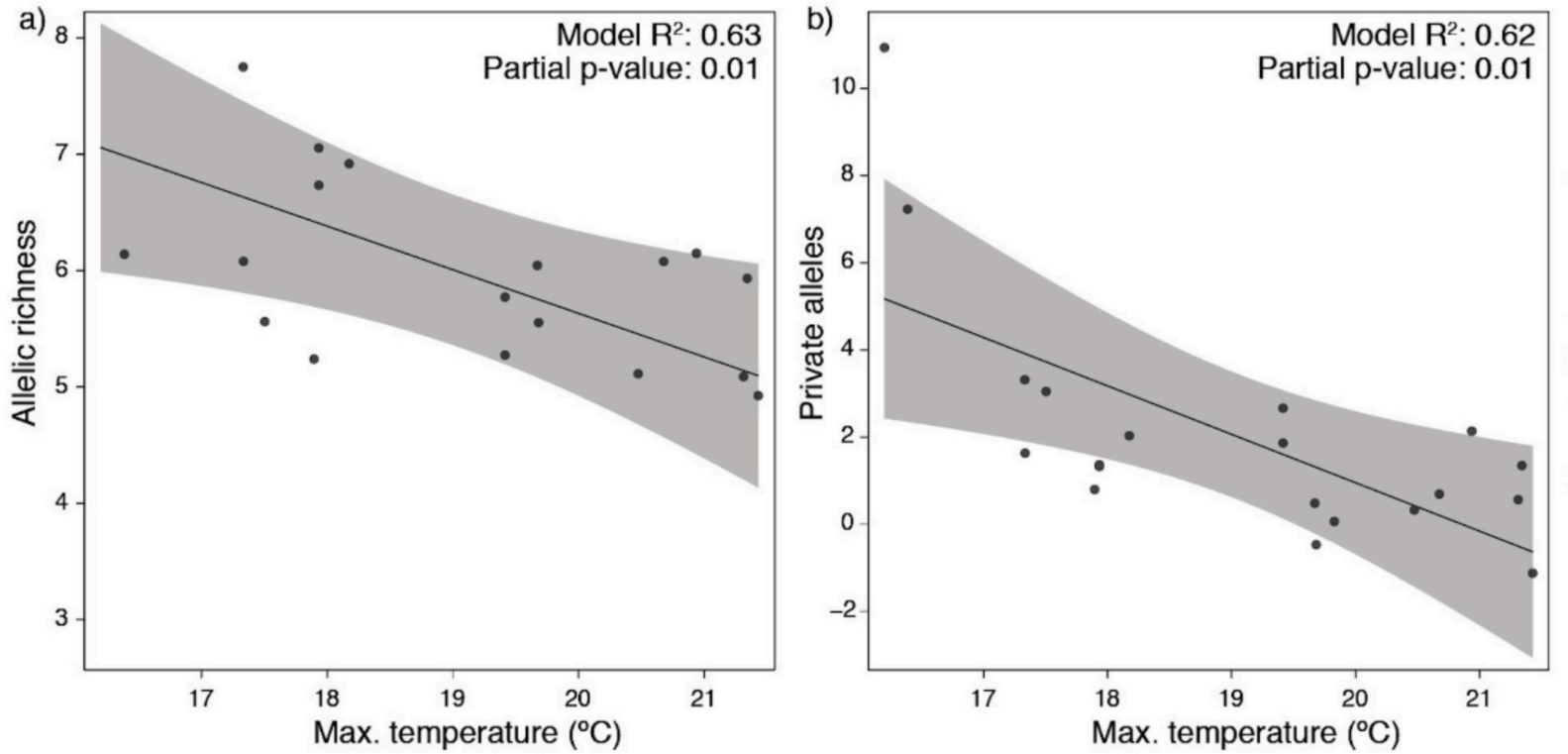
Sea-whip coral





Comparing shadow vs. deeper sites, we found **higher diversity in deeper populations.**

Potential erosion of shallow diversity due to warmer conditions (one exception).



A relationship of reduced diversity with warmer conditions.

Coral populations exposed to detrimental surface conditions showed reduced diversity.

Deep, colder environments can provide refugia for biodiversity.

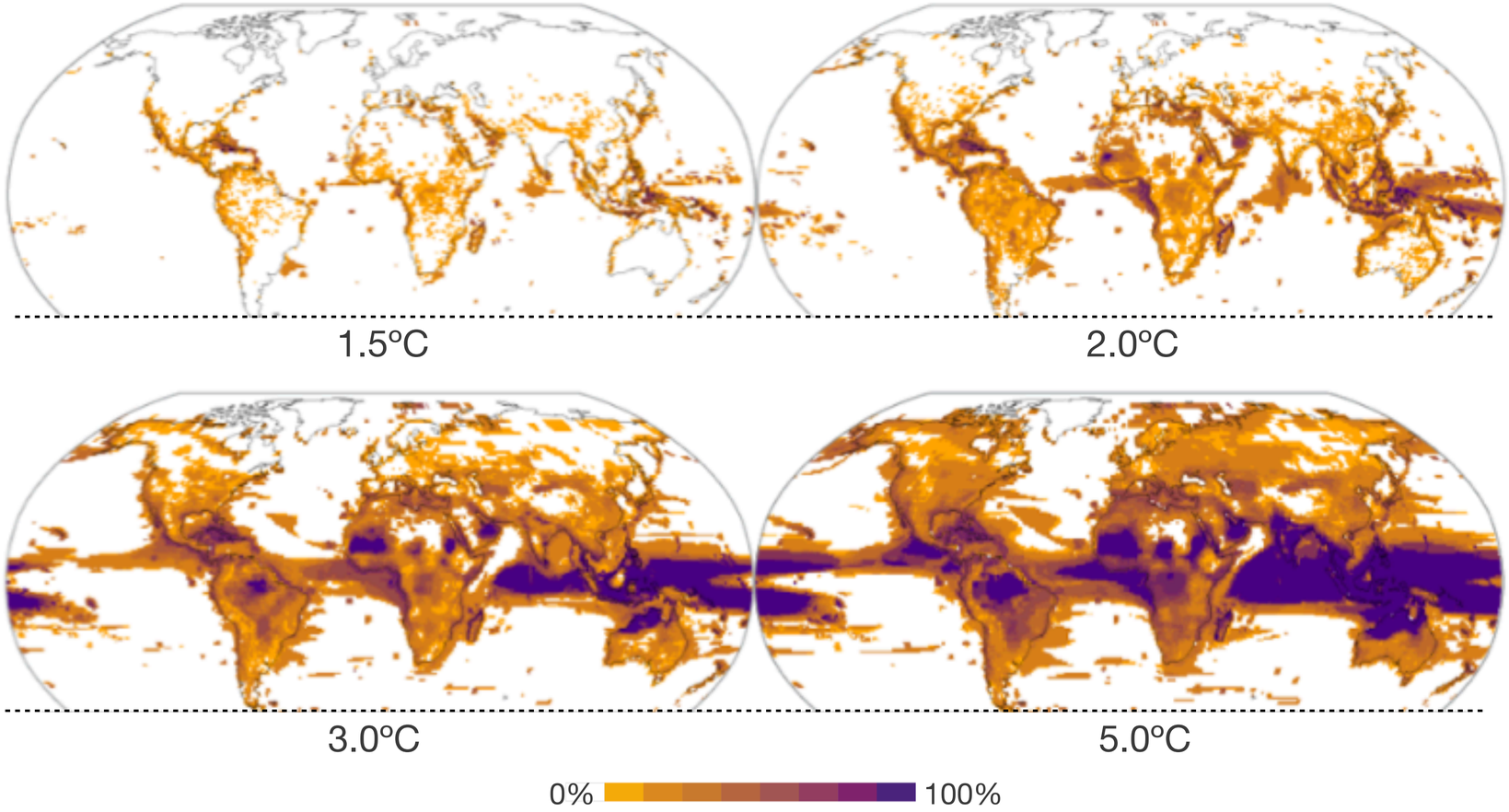
Climate change impacts can be amplified if elicited on ecosystem structuring species. Their loss or altered state can lead to the **simplification of ecosystem functioning and services****, such as blue carbon sequestration, habitats provisioning for associated species and shoreline protection.

** even when species are replaced by others with similar ecological roles.

Despite the observed climate-induced impacts on biodiversity, a **comprehensive impact assessment focused on future scenarios (e.g., IPCC) is still missing**.

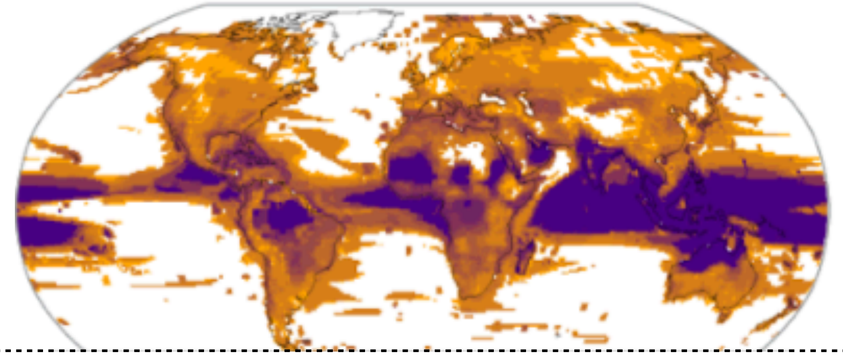
Projections limited to few species and regions, **none at the genetic diversity level.**

Machine learning species distribution modelling can forecast distribution changes under different climate change scenarios.



IPCC shows species exposed to detrimental conditions under increasing levels of warming (% of species affected), particularly at equatorial regions. Highlights the **importance of broad compliance with the Paris Agreement**.

IPCC, 2023



% species exposed to detrimental conditions

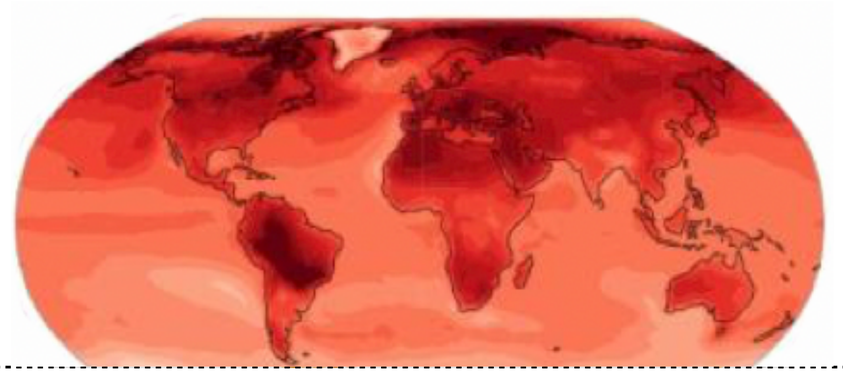
The report is quite **comprehensive, covering marine species** of fish, benthic invertebrates, plankton, cephalopods, corals and seagrasses.

Yet, a unique overall view of changes for all groups.

What are the patterns of the different groups and the drivers of change?

Will deep coral responses be on the same extent as those of coastal macroalgae?

IPCC, 2023

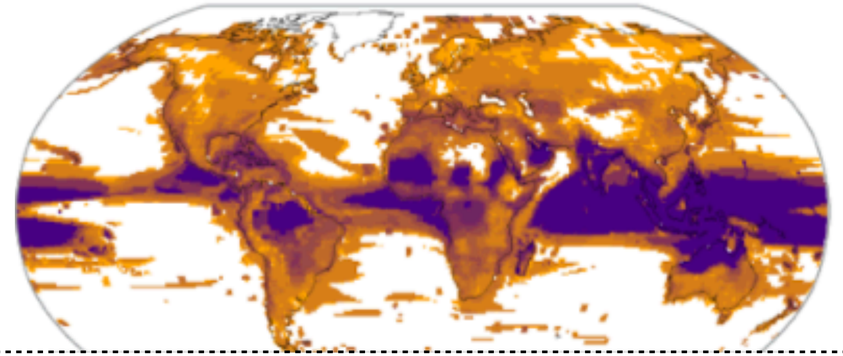


Global warming levels

Analyses based on temperatures alone, but **marine organisms** will likely respond to **additional drivers that may change in the future**.
(e.g., oxygen and primary productivity).

Only surface conditions considered, **overlooking the potential role of depth shifts and depth refugia,** where populations might persist away from the warming trends.

IPCC, 2023



% species exposed to detrimental conditions

Only **highlighting the potential drastic effects**, even of the optimistic climate scenarios.

What about opportunities for conservation, like refugia and hotspots of biodiversity?

We step into the **post-2020 global biodiversity framework**, a cornerstone of conservation endorsing the **protection 30% of the oceans through a climatically resilient** network of MPAs with **focus on important areas of biodiversity**.

IPCC, 2023

Despite the observed climate-induced impacts on biodiversity, a **comprehensive impact assessment focused on future scenarios (e.g., IPCC) is still missing**.

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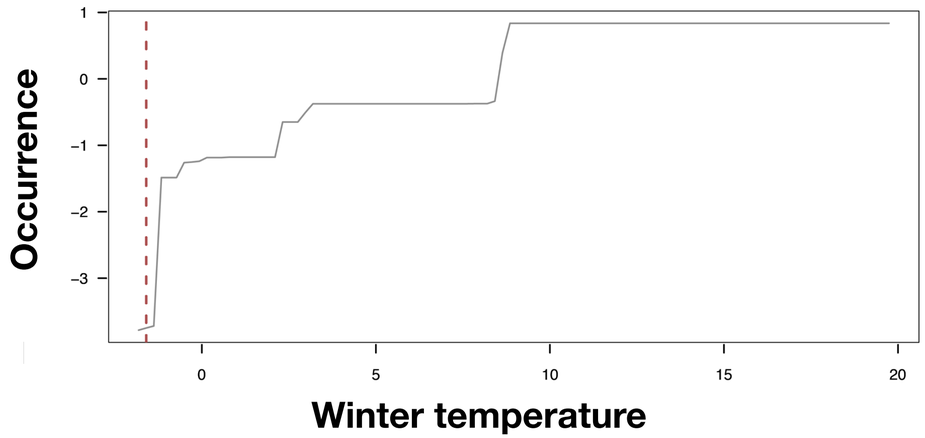
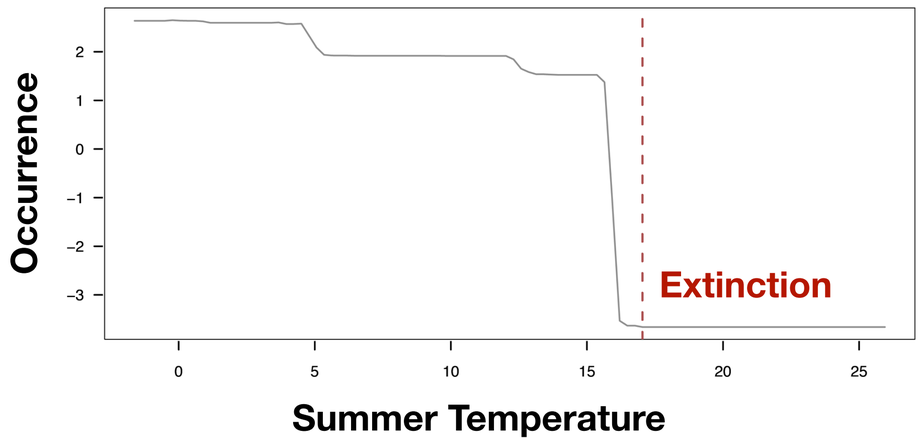
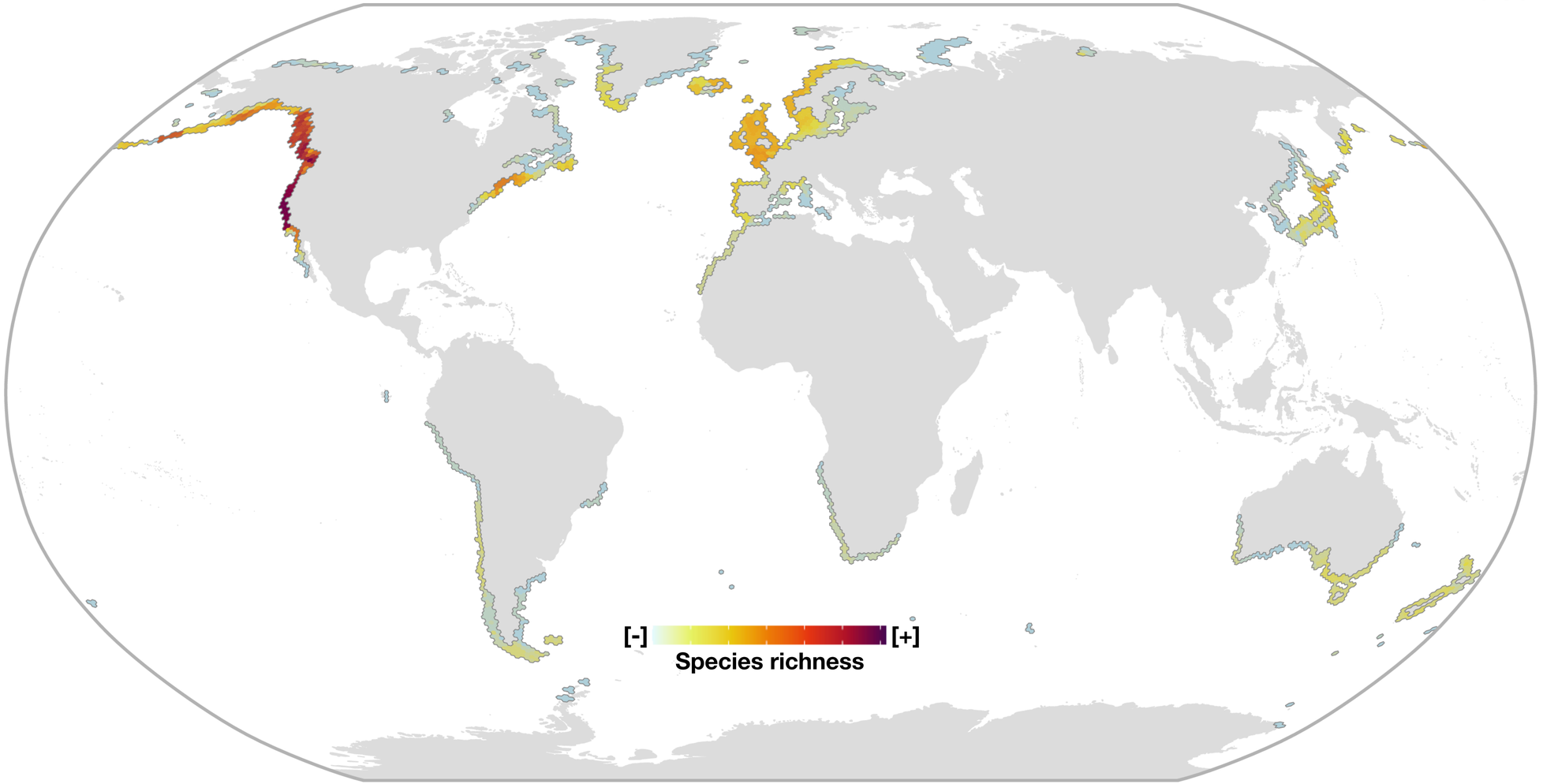


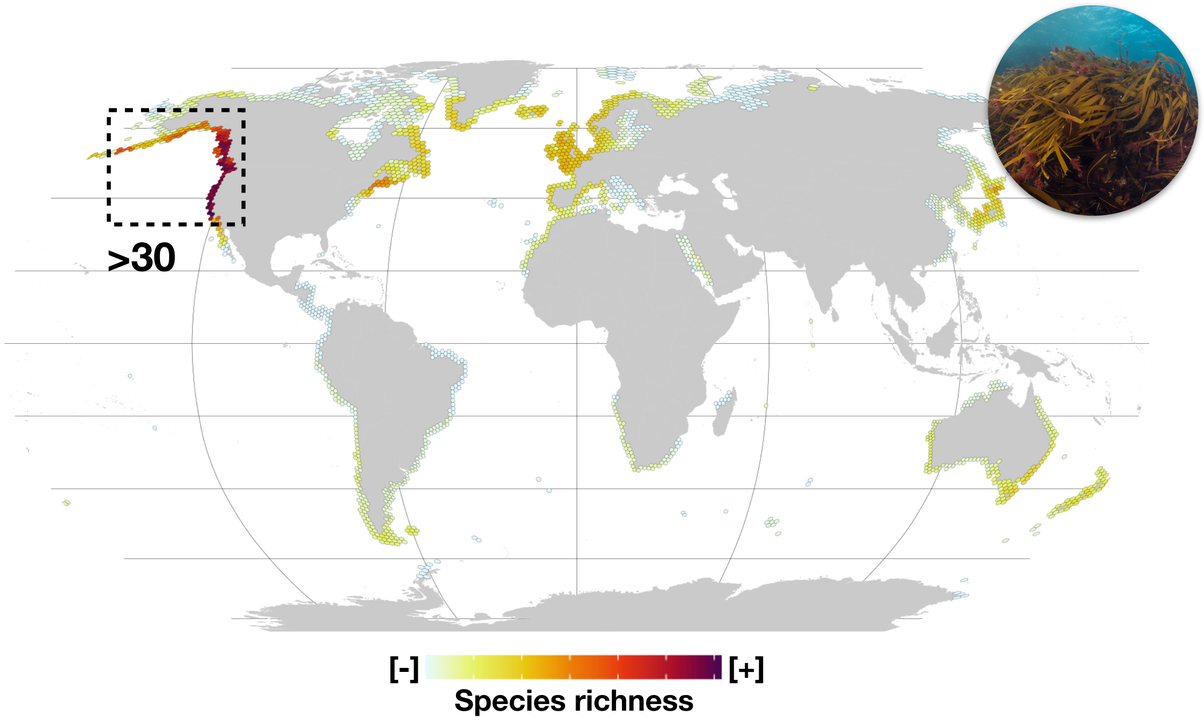


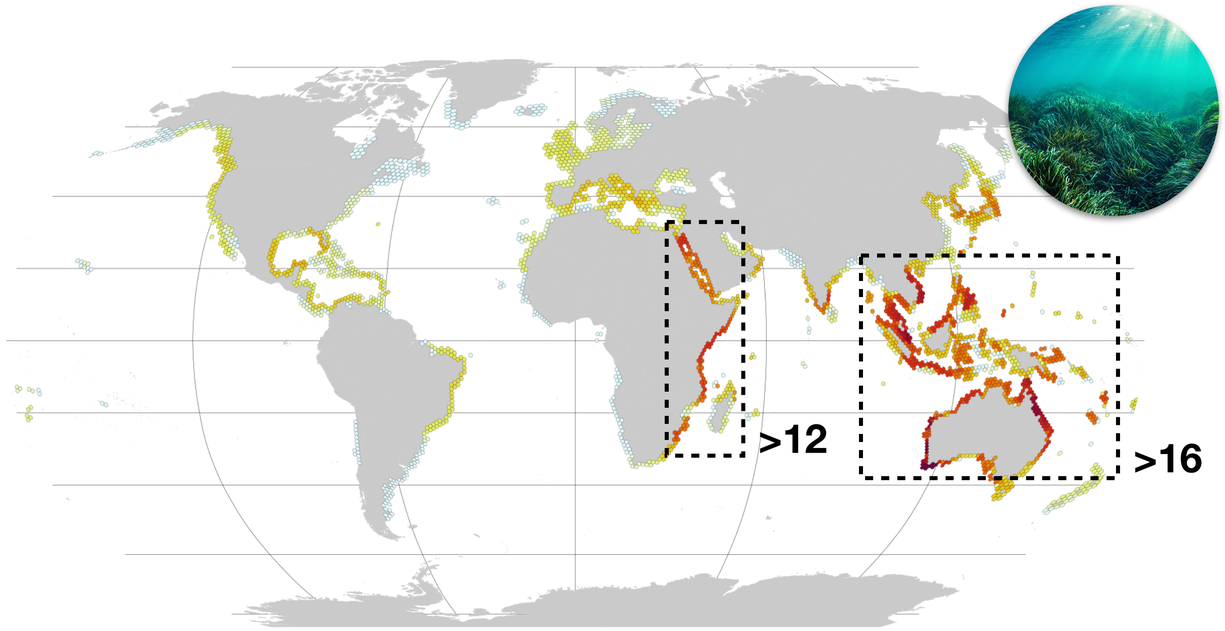
Predictive model

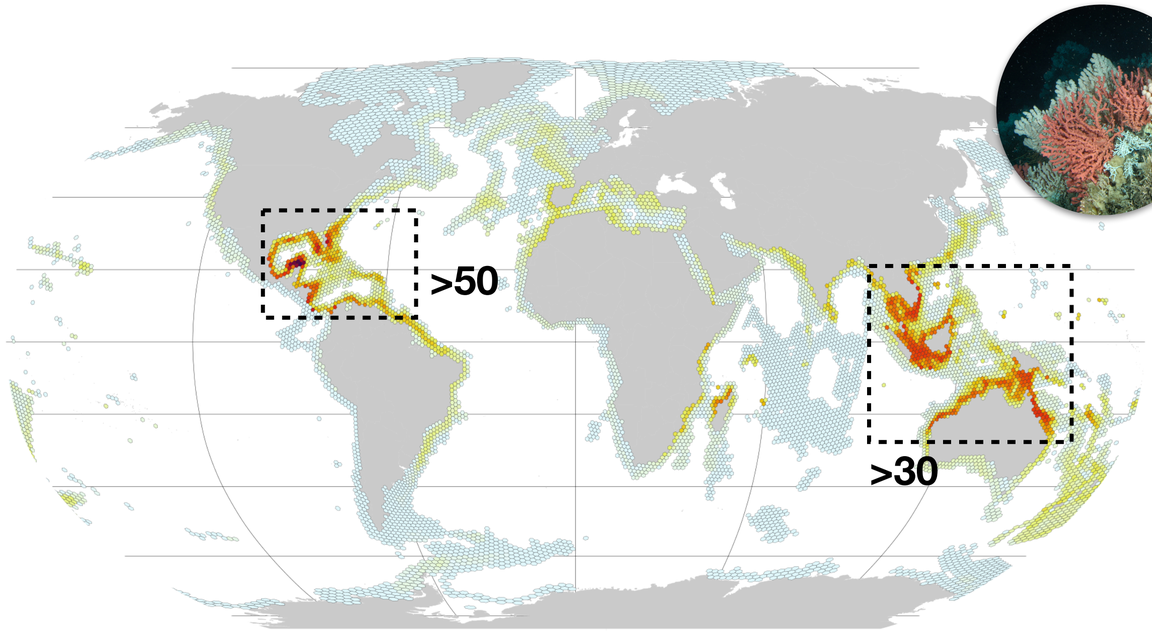
Distribution models (e.g., A.I.) **learn the effect of environmental variables on the distribution of species**, allowing to **predict their occurrence in space and time**, generating **continuous maps of habitat suitability**.



Stacking numerous individuals models we can predict species' richness and community composition, as well as **project biodiversity changes** (e.g., **community turnover**).





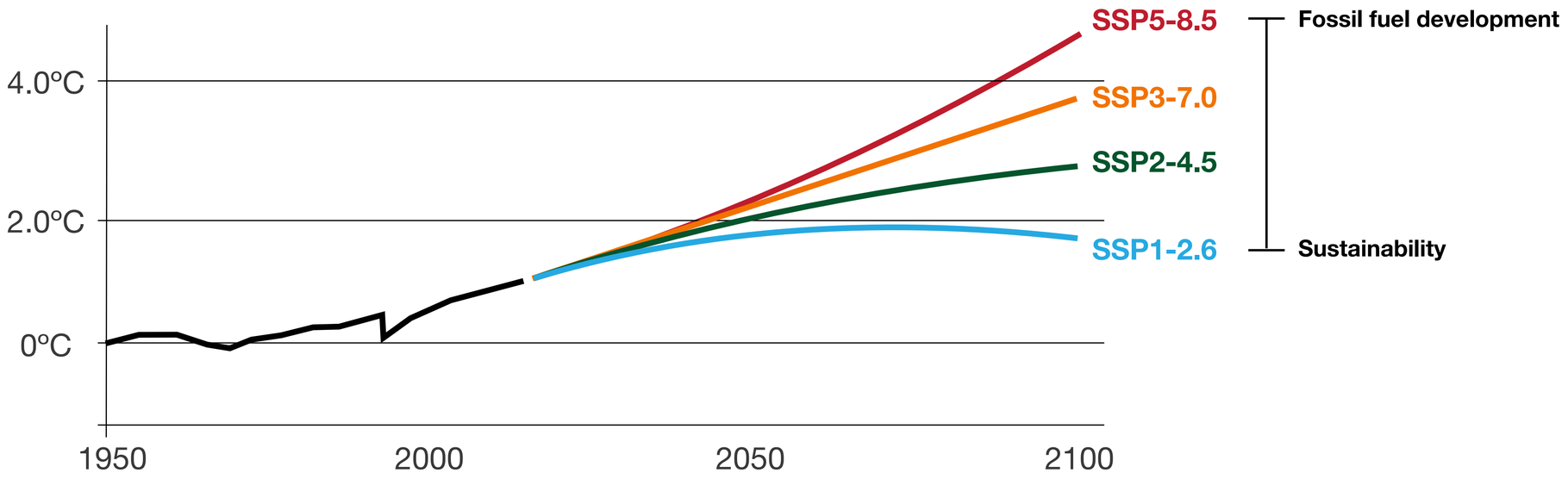


Stacked SDMs provided an overall view of species richness hotspots. Different patterns between groups

NE Pacific, from Point Conception to the Aleutians;

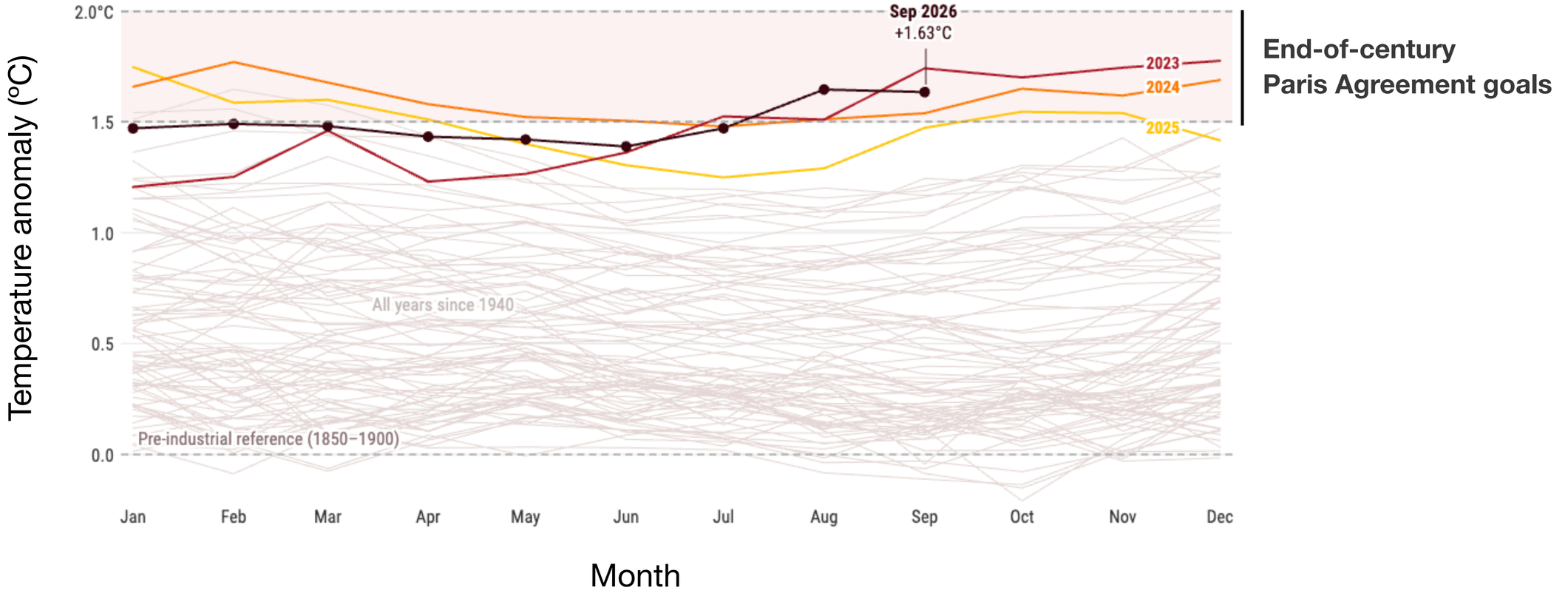
Indo-West Pacific, Australia and the Red Sea;

Caribbean Sea, coral triangle, and N Australia.



Projections made with the SSP scenarios, which explore **different climate trajectories**.

From sustainable socioeconomic trajectories, under the Paris Agreement expectations (warming < 2ºC), **to fossil fuel development** (much higher temperature changes).

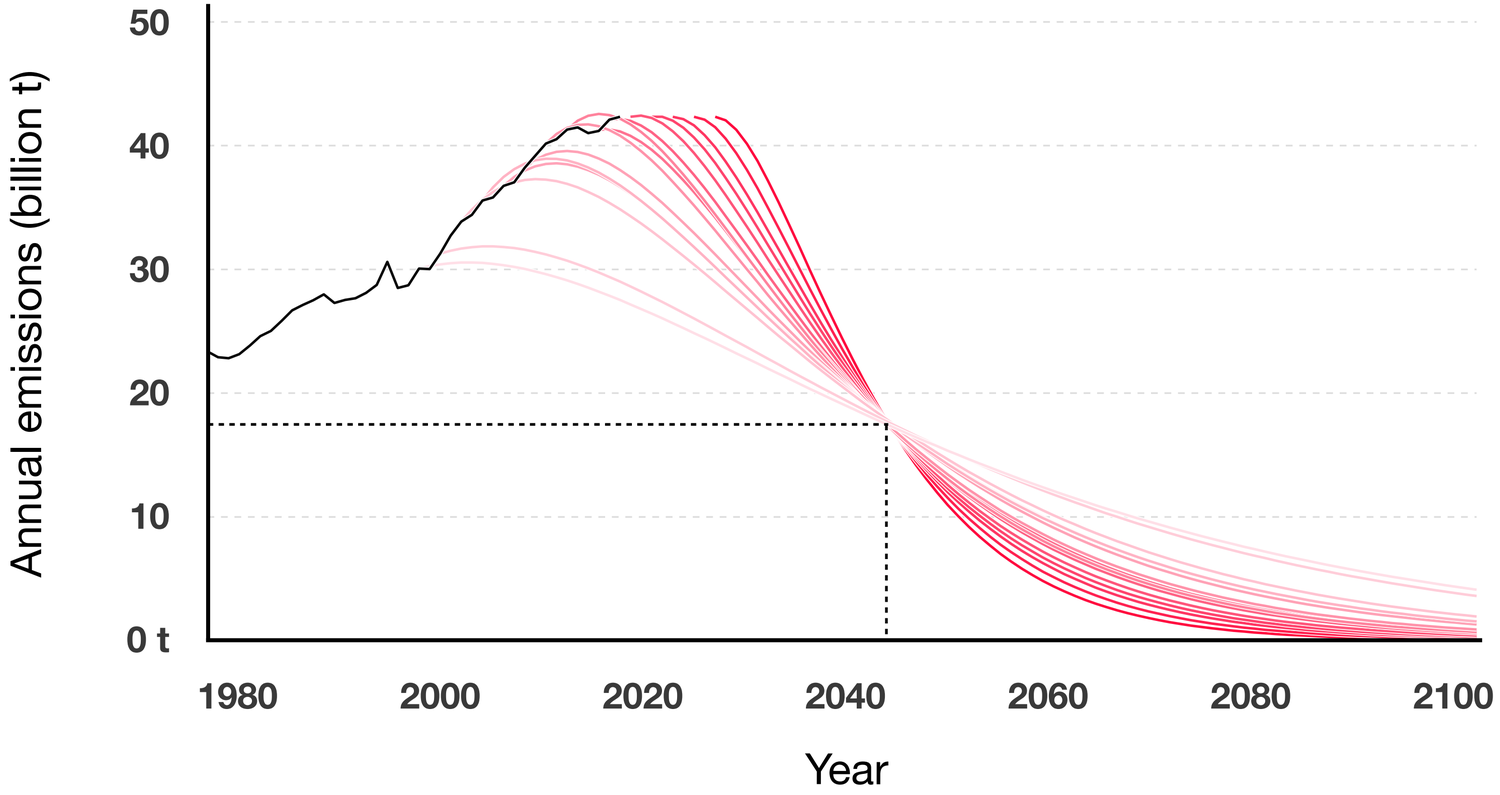


What is our current trajectory of climate change?

We are already surpassing the targets set by the Paris Agreement for end-of-century.

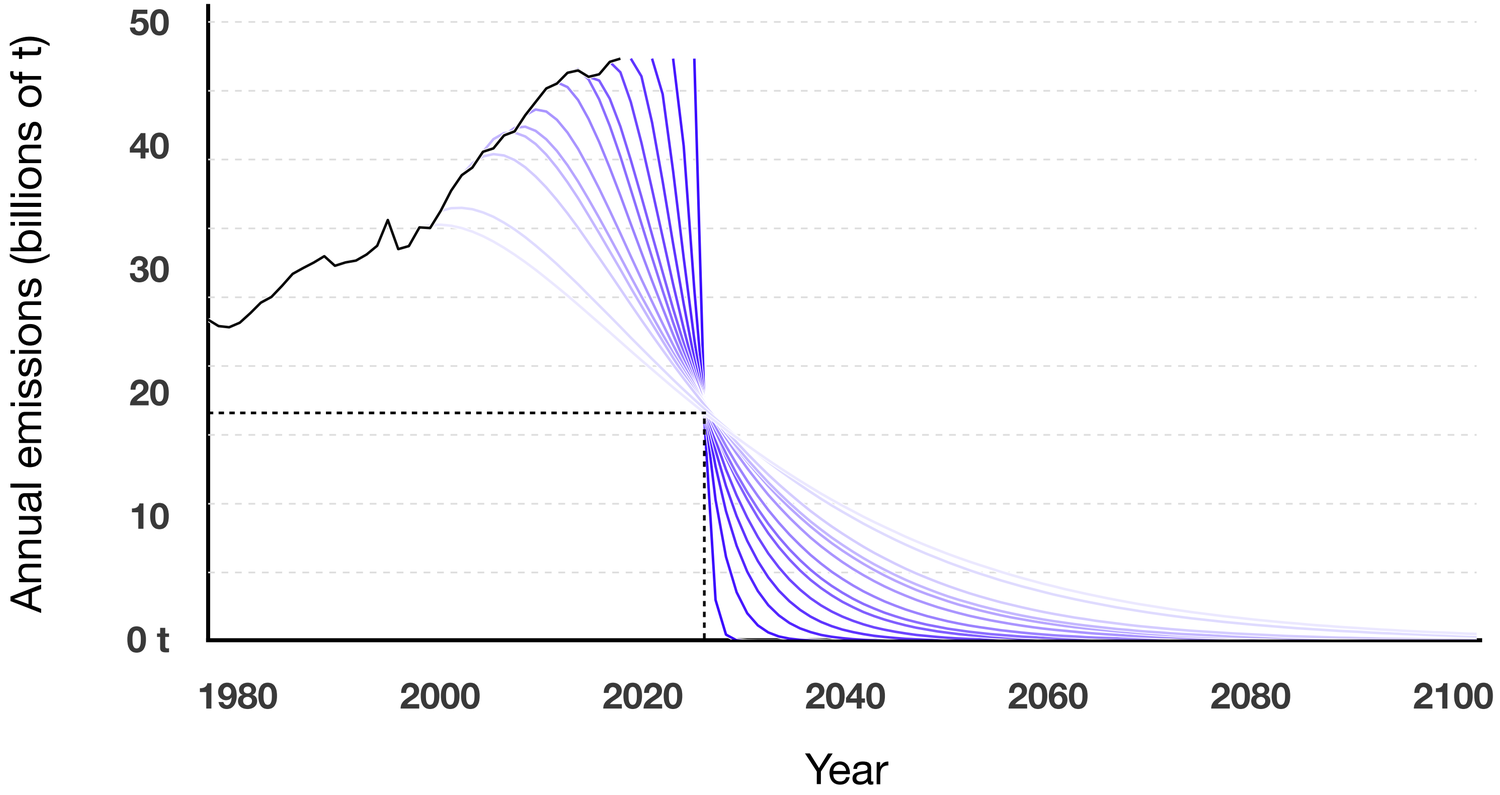
September 2026 was the 2nd warmest September ever recorded and marked the second consecutive month in 2026, with global temperatures exceeding 1.5°C.

Copernicus, 2026



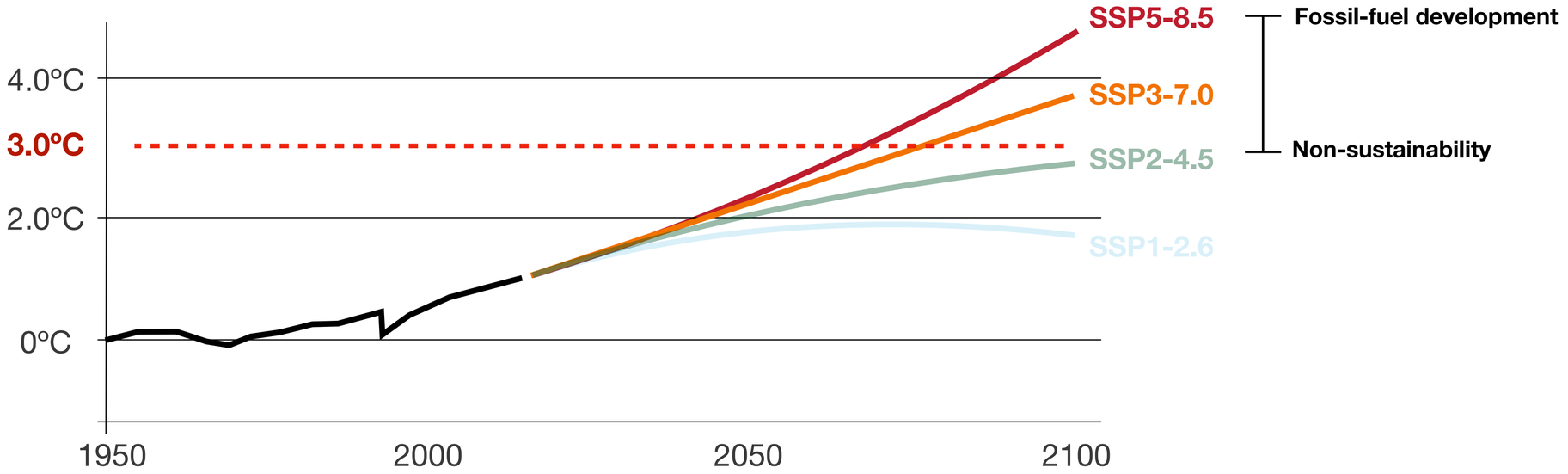
To **achieve the goal** of keeping global **temperatures below 2°C**, **reductions of greenhouse gas emissions would need to lower to 1970s levels by 2045**.

Copernicus, 2025



To **achieve the goal** of keeping global **temperatures below 1.5°C**, **reductions of greenhouse gas emissions would need to lower to 1970s levels by 2025**.

Copernicus, 2025



Climate researchers are **leaving behind SSP1-2 of the sustainability**.

SSP3 and above is **anticipated to produce major biodiversity shifts**.

2 min. PITCH

Papers: P47, P53, P54

10 minutes

Read the abstract and the introduction and interpret the main figures.

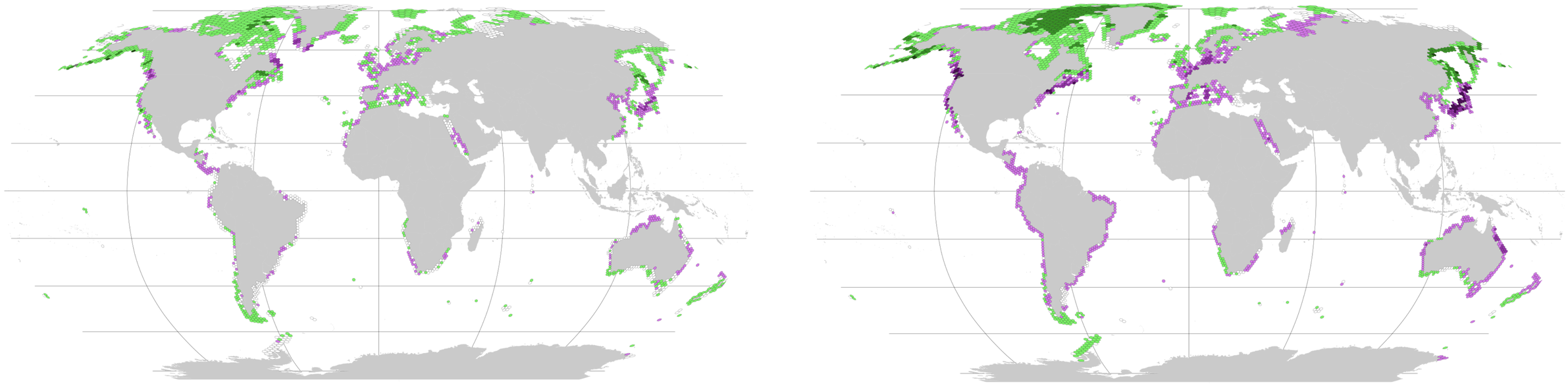
Question. What problem did the authors tackle? (one sentence)
Approach. What did they do, and with which data and kind of model? (one sentence)
Key finding. The one result to remember, ideally with the paper's main figure.
Why it matters today?

Low carbon emissions

SSP1-1.9

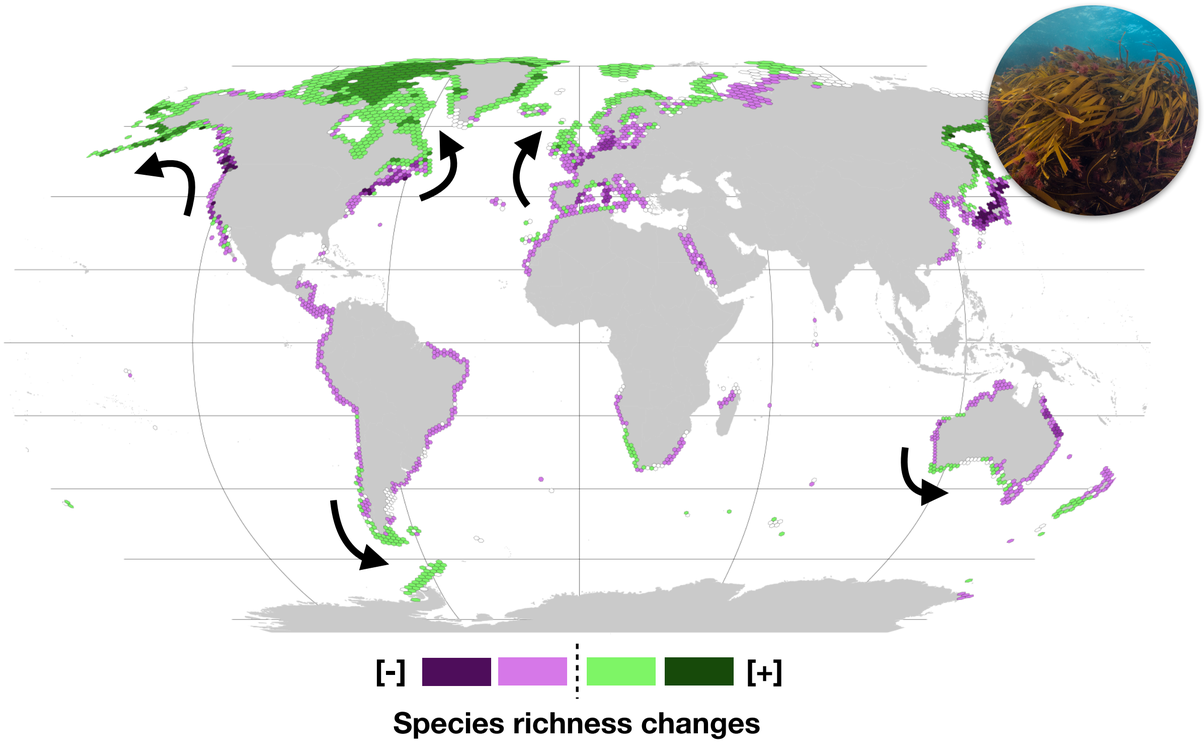
High carbon emissions

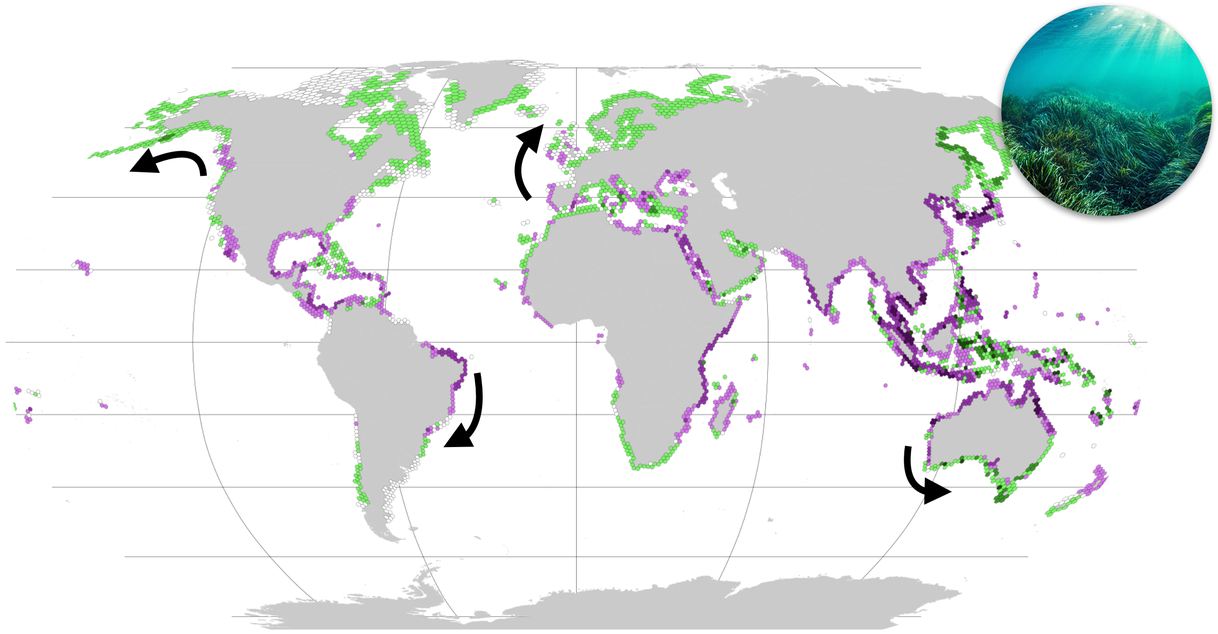
SSP3-7.0

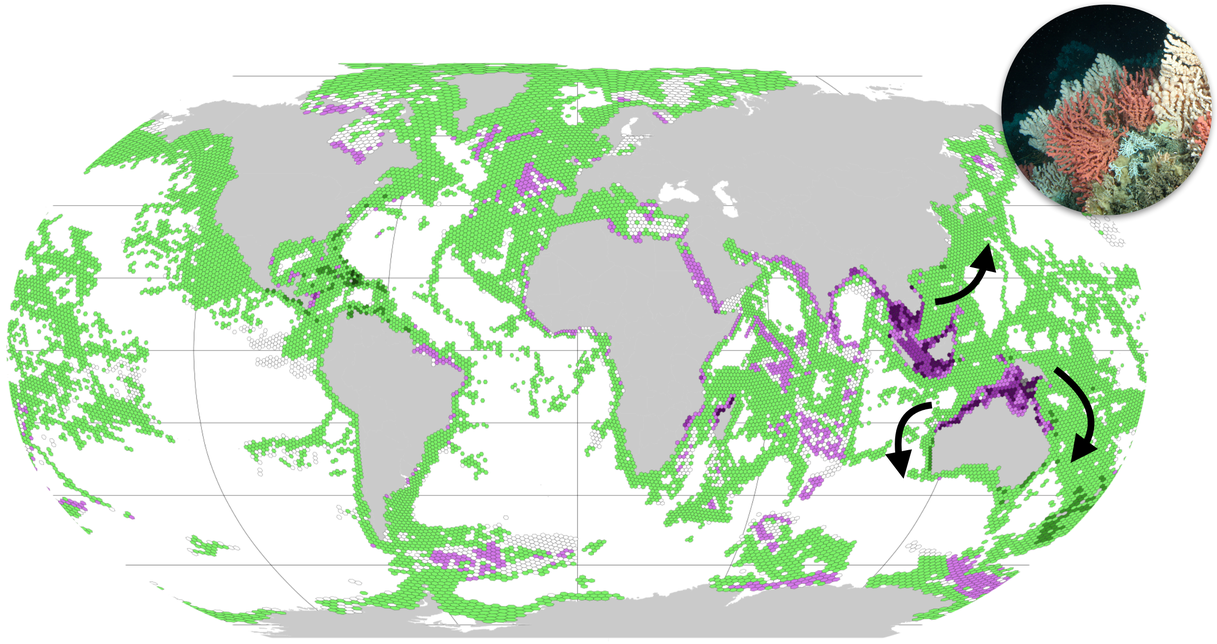


Projections showed that, **regardless of the group, higher emissions are link to increased species richness changes due to poleward shifts.**

Assis et al., 2024







Species richness changes

More emissions, more change:

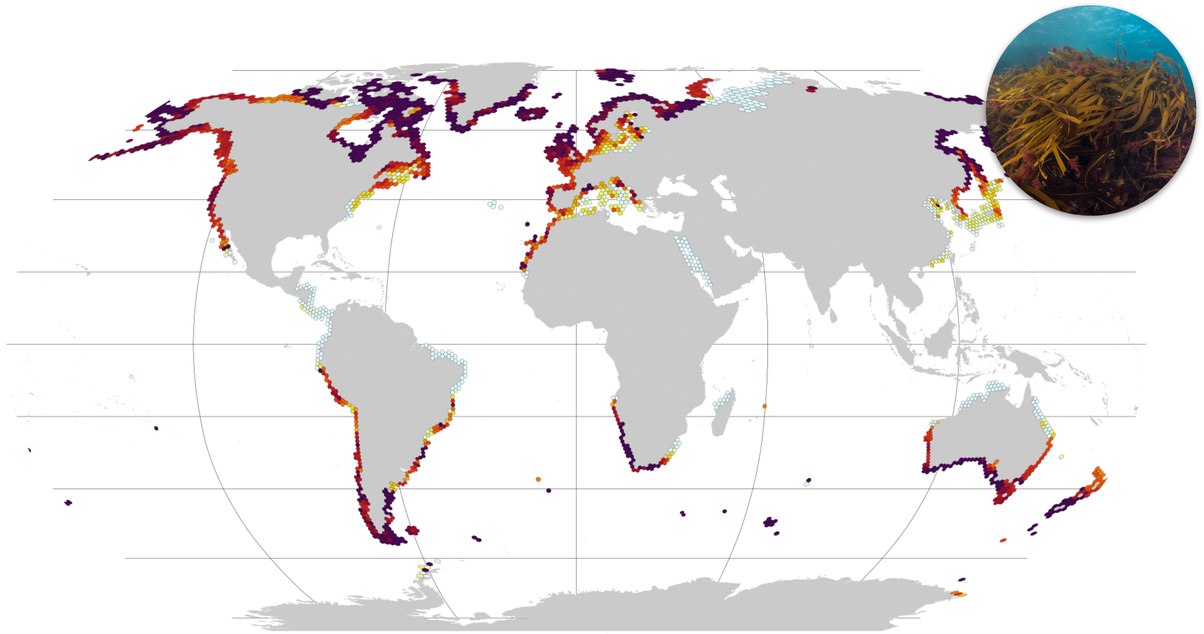
Poleward range shifts producing diversity losses at equatorial regions and increases at higher latitudes. Species' **replacements at climatic transition zones**.

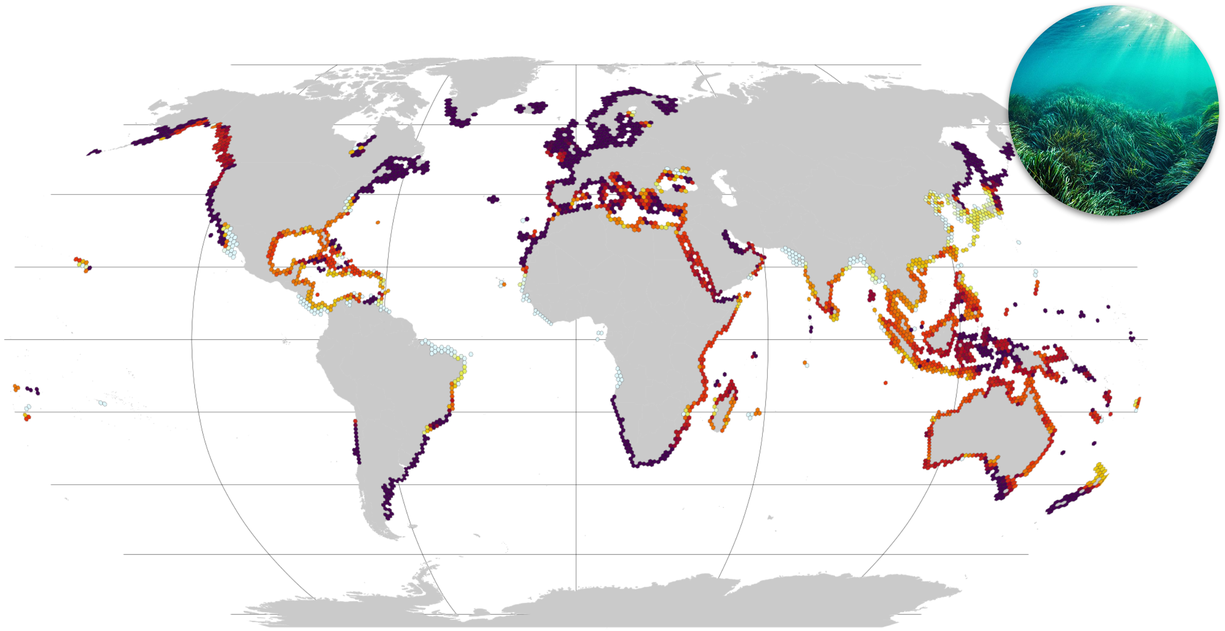
Poleward range shifts from the Indo-West Pacific (coral triangle, and N Australia), **depth shifts globally.**

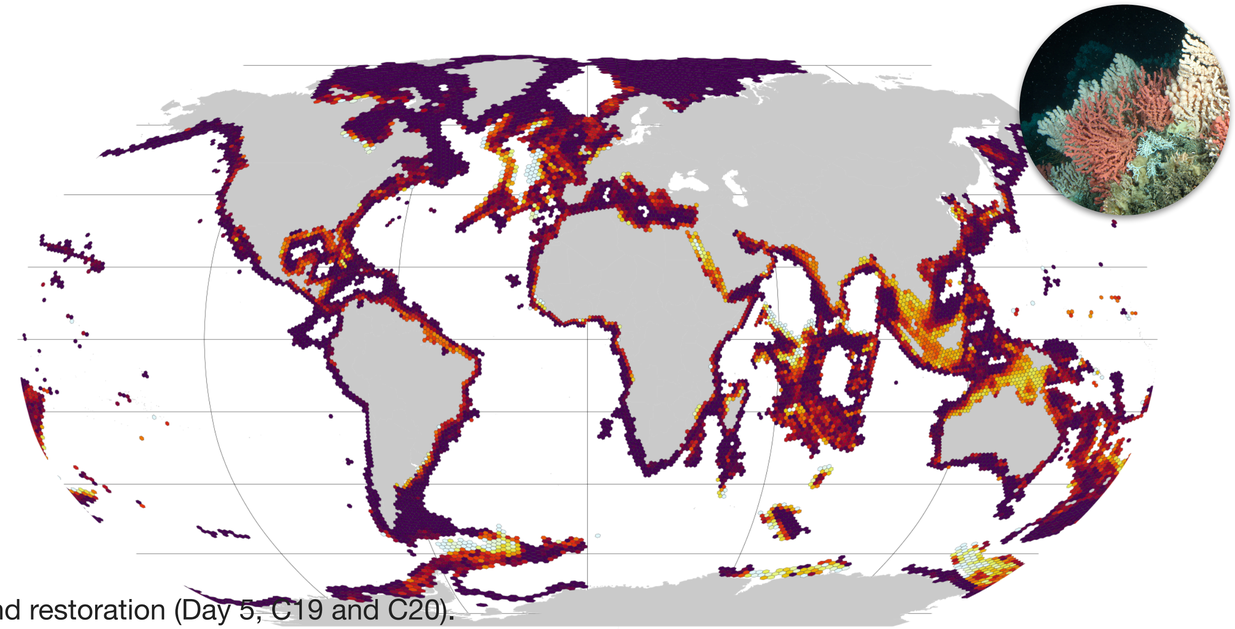
Fragkopoulou et al, 2025

Gouvêa et al, 2024

Assis et al, 2024







Refugia (providing long-term persistence)

High degree of refugia, globally and in depth:

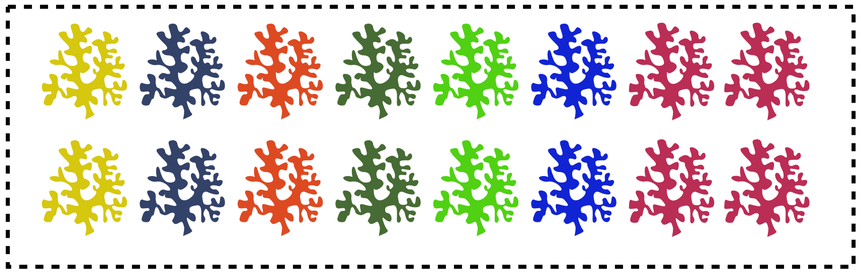
- Higher latitudes [+ 0.57 deep];
- Higher latitudes + Eastern coral triangle [+ 1.59 deep];
- Globally, offshore deeper areas [+ 6.75 deep];

From maps to decisions: refugia and threatened regions can inform conservation, management and restoration (Day 5, C19 and C20).

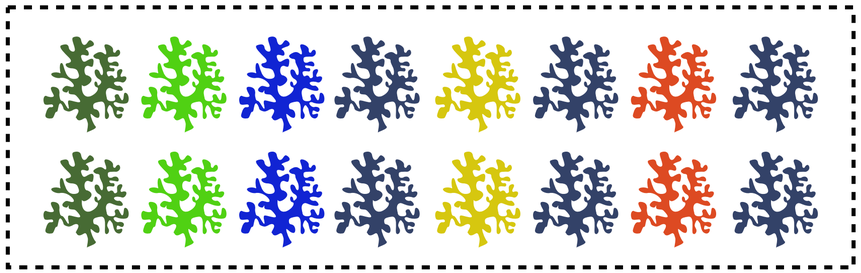
Gouvêa et al, 2024

Assis et al, 2024

Population 1



Population 2

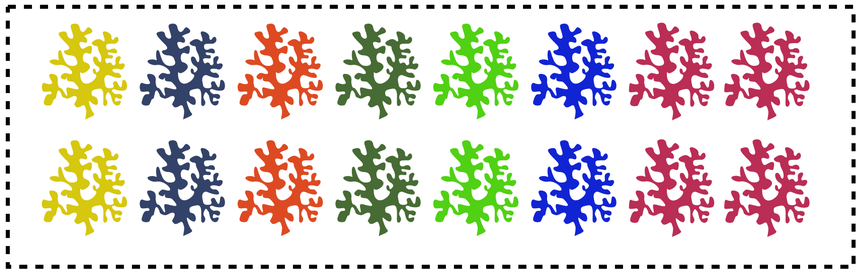


The Hidden Loss: Genetic Consequences of Climate Change

Population 1

Temperature within phys. limits

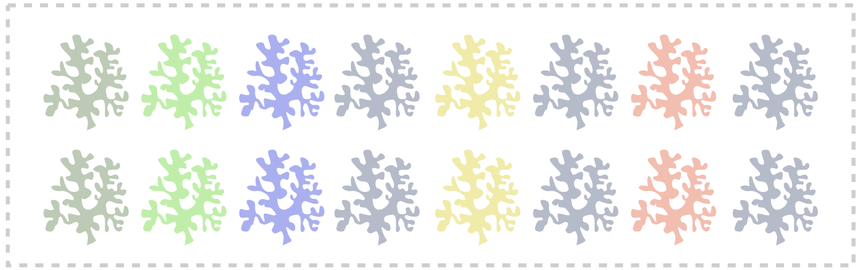
Climatic refugia



Population 2

Temperature above limits

Local extinction, loss of diversity

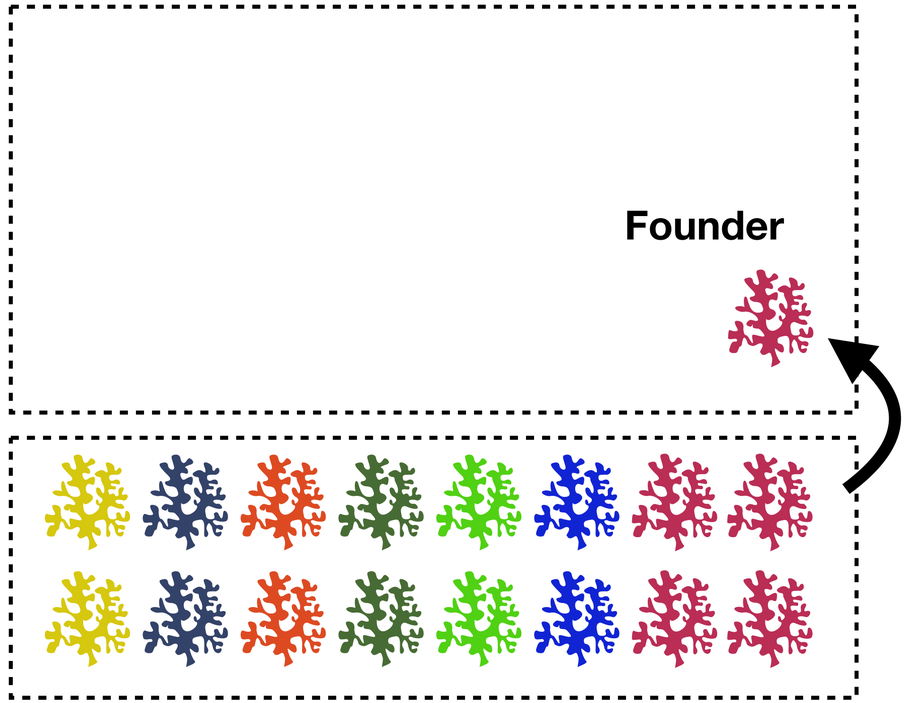


Warming

The Hidden Loss: Genetic Consequences of Climate Change

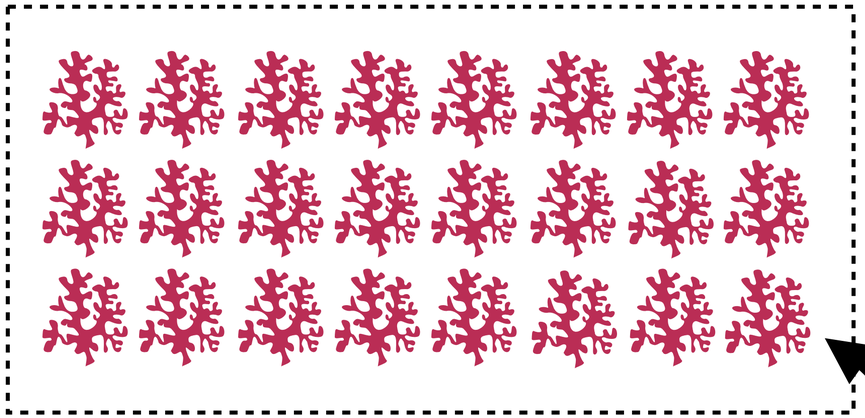
Population 3 (new)
Temperature tolerated
Demographic expansion

Population 1
Temperature within phys. limits
Climatic refugia

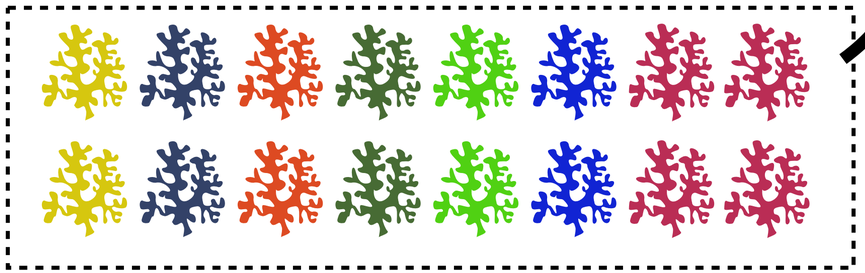


The Hidden Loss: Genetic Consequences of Climate Change

Population 3 (new)
Temperature tolerated
Demographic expansion



Population 1
Temperature tolerated
Climatic refugia



Warming

The Hidden Loss: Genetic Consequences of Climate Change

Why we need to model the distribution of biodiversity?

We are **observing massive changes owing to ongoing climate change; and we need to fulfill climate and conservation policy agendas**.

IPCC reports and the Paris Agreement: Projections provide relevant climate change impact assessments, which highlight the role of the Paris Agreement, with **increased emissions linked to the global restructure ecosystems**, impacting their functioning and services - ***where and in what extent are projected changes?***

Post-2020 Global Biodiversity Framework of the Convention on Biological Diversity: Models identify conservation opportunities - ***where are the long-term climatic refugia and hotspots of biodiversity?*** This sites can be subject to higher conservation efforts (e.g., implementation of PA).